

Question #1 of 164

Question ID: 1479903

When constructing a regression model to predict portfolio returns, an analyst runs a regression for the past five year period. After examining the results, she determines that an increase in interest rates two years ago had a significant impact on portfolio results for the time of the increase until the present. By performing a regression over two separate time periods, the analyst would be attempting to prevent which type of misspecification?

A) Incorrectly pooling data.



B) Inappropriate variable scaling.



C) Inappropriate variable form.



Explanation

The relationship between returns and the dependent variables can change over time, so it is critical that the data be pooled correctly. Running the regression for multiple sub-periods (in this case two) rather than one time period can produce more accurate results.

(Module 1.3, LOS 1.g)

Phillip Lee works for Song Bank as a quantitative analyst. He is currently working on a model to explain the returns (in %) of 20 hedge funds for the past year. He includes three independent variables:

- Market return = return on a broad-based stock index (in %)
- Closed = dummy variable (= 1 if the fund is closed to new investors; 0 otherwise)
- Prior period alpha = fund return for the prior 12 months – return on market (in %)

Estimated model: hedge fund return = $3.2 + 0.22 \text{ market return} + 1.65 \text{ closed} - 0.11 \text{ prior period alpha}$

Lee is concerned about the impact of outliers on the estimated regression model and collects the following information:

Observation	1	2	3	4	5	6	7	8	9	10
Cook's D	0.332	0.219	0.115	0.212	0.376	0.232	0.001	0.001	0.233	0.389
Observation	11	12	13	14	15	16	17	18	19	20
Cook's D	0.089	0.112	0.001	0.001	0.219	0.001	0.112	0.044	0.517	0.212

Additionally, Lee wants to estimate the probability of a hedge fund closing to new investors, and he uses two variables:

- Fund size = log of assets under management
- Prior period alpha (defined earlier)

Results are shown as follows:

Variable	Coefficient
Intercept	-3.76
Fund size	-2.98
Prior period alpha	-2.99

Question #2 - 5 of 164

Question ID: 1479961

What is the correct interpretation of the coefficient of *closed* in the first regression?

- A)** A closed fund is estimated to have an extra return of 1.65% relative to funds that are not closed. 
- B)** A closed fund is likely to generate a return of 1.65%. 
- C)** If a model is closed to new investors, the expected excess fund return is 1.65%. 

Explanation

The interpretation of the coefficient is the extra return relative to the alternative outcome.

(Module 1.4, LOS 1.I)

Question #3 - 5 of 164

Question ID: 1479962

To check for only the outliers in the sample, Lee should *most appropriately* use:

- A)** Cook's D. 
- B)** Studentized residuals. 
- C)** leverage. 

Explanation

Studentized residuals are used to identify outliers (in the dependent variable). Leverage is used to identify high-leverage observations (in the independent variable), while Cook's D is a composite measure (combines both independent and dependent variables) to identify influential observations.

(Module 1.4, LOS 1.k)

Question #4 - 5 of 164

Question ID: 1479963

Which observations, when excluded, cause a significant change to model coefficients?

- A) Observation 19. 
- B) Observations 1, 10, and 11. 
- C) Observations 10 and 19. 

Explanation

Influential observations are those that, when excluded, cause a significant change to the model coefficients.

$$\text{Observations where Cook's } D > \sqrt{\frac{k}{n}} = \sqrt{\frac{3}{20}} = 0.3873.$$

Observations 10 (D = 0.389) and 19 (D = 0.517) satisfy this criteria.

(Module 1.4, LOS 1.k)

Question #5 - 5 of 164

Question ID: 1479964

What is the change in probability of fund closure for a 1% increase in Ln(assets under management)?

- A) 4.83%. 
- B) 2.33%. 
- C) 5.08%. 

Explanation

$$\text{Odds} = e^{\text{coeff (fund size)}} = e^{-2.98} = 0.0508.$$

$$\text{Probability} = \text{odds} / (1 + \text{odds}) = 0.0483 = 4.83\%.$$

(Module 1.4, LOS 1.m)

Question #6 of 164

Question ID: 1479907

Which of the following conditions will *least likely* affect the statistical inference about regression parameters by itself?

A) Unconditional heteroskedasticity.



B) Multicollinearity.



C) Model misspecification.



Explanation

Unconditional heteroskedasticity does not impact the statistical inference concerning the parameters. Misspecified models have inconsistent and biased regression parameters. Multicollinearity results in unreliable estimates of regression parameters.

(Module 1.3, LOS 1.h)

(Module 1.3, LOS 1.i)

(Module 1.3, LOS 1.j)

Question #7 of 164

Question ID: 1479959

A high-yield bond analyst is trying to develop an equation using financial ratios to estimate the probability of a company defaulting on its bonds. A technique that can be used to develop this equation is:

A) dummy variable regression.



B) logistic regression model.



C) multiple linear regression adjusting for heteroskedasticity.



Explanation

The only one of the possible answers that estimates a probability of a discrete outcome is logit or logistic modeling.

(Module 1.4, LOS 1.m)

Question #8 of 164

Question ID: 1472074

The management of a large restaurant chain believes that revenue growth is dependent upon the month of the year. Using a standard 12 month calendar, how many dummy variables must be used in a regression model that will test whether revenue growth differs by month?

A) 11.



B) 12.



C) 13.



Explanation

The appropriate number of dummy variables is one less than the number of categories because the intercept captures the effect of the other effect. With 12 categories (months) the appropriate number of dummy variables is $11 = 12 - 1$. If the number of dummy variables equals the number of categories, it is possible to state any one of the independent dummy variables in terms of the others. This is a violation of the assumption of the multiple linear regression model that none of the independent variables are linearly related.

(Module 1.4, LOS 1.I)

Question #9 of 164

Question ID: 1479914

Which of the following questions is *least likely* answered by using a qualitative dependent variable?

A) Based on the following company-specific financial ratios, will company ABC enter bankruptcy?



B) Based on the following executive-specific and company-specific variables, how many shares will be acquired through the exercise of executive stock options?



C) Based on the following subsidiary and competition variables, will company XYZ divest itself of a subsidiary?



Explanation

The number of shares can be a broad range of values and is, therefore, not considered a qualitative dependent variable.

(Module 1.3, LOS 1.h)

Question #10 of 164

Question ID: 1479934

A regression with three independent variables have VIF values of 3, 4, and 2 for the first, second, and third independent variables, respectively. Which of the following conclusions is *most* appropriate?

- A) Total VIF of 9 indicates a serious multicollinearity problem. 
- B) Only variable two has a problem with multicollinearity. 
- C) Multicollinearity does not seem to be a problem with the model. 

Explanation

Multicollinearity occurs when an independent variable is highly correlated with a linear combination of the remaining independent variables. VIF values exceeding 5 need to be investigated while values exceeding 10 indicate strong evidence of multicollinearity.

(Module 1.3, LOS 1.j)

Question #11 of 164

Question ID: 1471982

Consider the following estimated regression equation, with calculated t -statistics of the estimates as indicated:

$$\text{AUTO}_t = 10.0 + 1.25 \text{PI}_t + 1.0 \text{TEEN}_t - 2.0 \text{INS}_t$$

with a PI calculated t -statistic of 0.45, a TEEN calculated t -statistic of 2.2, and an INS calculated t -statistic of 0.63.

The equation was estimated over 40 companies. The predicted value of AUTO if PI is 4, TEEN is 0.30, and INS = 0.6 is *closest* to:

- A) 17.50. 
- B) 14.10. 
- C) 14.90. 

Explanation

Predicted AUTO

$$\begin{aligned} &= 10 + 1.25 (4) + 1.0 (0.30) - 2.0 (0.6) \\ &= 10 + 5 + 0.3 - 1.2 \\ &= 14.10 \end{aligned}$$

(Module 1.2, LOS 1.f)

Toni Williams, CFA, has determined that commercial electric generator sales in the Midwest U.S. for Self-Start Company is a function of several factors in each area: the cost of heating oil, the temperature, snowfall, and housing starts. Using data for the most currently available year, she runs a cross-sectional regression where she regresses the deviation of sales from the historical average in each area on the deviation of each explanatory variable from the historical average of that variable for that location. She feels this is the most appropriate method since each geographic area will have different average values for the inputs, and the model can explain how current conditions explain how generator sales are higher or lower from the historical average in each area. In summary, she regresses current sales for each area minus its respective historical average on the following variables for each area.

- The difference between the retail price of heating oil and its historical average.
- The mean number of degrees the temperature is below normal in Chicago.
- The amount of snowfall above the average.
- The percentage of housing starts above the average.

Williams used a sample of 26 observations obtained from 26 metropolitan areas in the Midwest U.S. The results are in the tables below. The dependent variable is in sales of generators in millions of dollars.

Coefficient Estimates Table		
Variable	Estimated Coefficient	Standard Error of the Coefficient
Intercept	5.00	1.850
\$ Heating Oil	2.00	0.827
Low Temperature	3.00	1.200
Snowfall	10.00	4.833
Housing Starts	5.00	2.333

Analysis of Variance Table (ANOVA)			
Source	Degrees of Freedom	Sum of Squares	Mean Square
Regression	4	335.20	83.80
Error	21	606.40	28.88
Total	25	941.60	

Table of the F-Distribution

Critical values for right-hand tail area equal to 0.05

Numerator: df1 and Denominator: df2

	df1				
df2	1	2	4	10	20
1	161.45	199.50	224.58	241.88	248.01
2	18.513	19.000	19.247	19.396	19.446
4	7.7086	6.9443	6.3882	5.9644	5.8025
10	4.9646	4.1028	3.4780	2.9782	2.7740
20	4.3512	3.4928	2.8661	2.3479	2.1242

One of her goals is to forecast the sales of the Chicago metropolitan area next year. For that area and for the upcoming year, Williams obtains the following projections: heating oil prices will be \$0.10 above average, the temperature in Chicago will be 5 degrees below normal, snowfall will be 3 inches above average, and housing starts will be 3% below average.

In addition to making forecasts and testing the significance of the estimated coefficients, she plans to perform diagnostic tests to verify the validity of the model's results.

Question #12 - 16 of 164

Question ID: 1471970

According to the model and the data for the Chicago metropolitan area, the forecast of generator sales is:

- A) \$35.2 million above the average.
- B) \$55 million above average.
- C) \$65 million above the average.



Explanation

The model uses a multiple regression equation to predict sales by multiplying the estimated coefficient by the observed value to get:

$$[5 + (2 \times 0.10) + (3 \times 5) + (10 \times 3) + (5 \times (-3))] \times \$1,000,000 = \$35.2 \text{ million.}$$

(Module 1.2, LOS 1.e)

Question #13 - 16 of 164

Question ID: 1479876

Williams proceeds to test the hypothesis that none of the independent variables has significant explanatory power. Using the joint F -test for the significance of all slope coefficients, at a 5% level of significance:

- A) at least one of the independent variables has explanatory power. 
- B) none of the independent variables has explanatory power. 
- C) all of the independent variables have explanatory power. 

Explanation

From the ANOVA table, the calculated F -statistic is (mean square regression / mean square error) = $(83.80 / 28.88) = 2.9017$. From the F distribution table (4 df numerator, 21 df denominator) the critical F value is 2.84. Because 2.9017 is greater than 2.84, Williams rejects the null hypothesis and concludes that at least one of the independent variables has explanatory power.

(Module 1.2, LOS 1.e)

Question #14 - 16 of 164

Question ID: 1479877

With respect to testing the validity of the model's results, Williams may wish to perform:

- A) a Breusch-Pagan test, but not Breusch-Godfrey. 
- B) both a Breusch-Godfrey test and a Breusch-Pagan test. 
- C) a Breusch-Godfrey test, but not a Breusch-Pagan test. 

Explanation

Since the model utilized is not an autoregressive time series, a test for serial correlation is appropriate so the Breusch-Godfrey test would be used. The Breusch-Pagan test for heteroskedasticity would also be a good idea.

(Module 1.2, LOS 1.e)

Question #15 - 16 of 164

Question ID: 1471974

When Williams ran the model, the computer said the R^2 is 0.233. She examines the other output and concludes that this is the:

- A) adjusted R^2 value. 
- B) neither the unadjusted nor adjusted R^2 value, nor the coefficient of correlation. 

C) unadjusted R^2 value.



Explanation

This can be answered by recognizing that the unadjusted R-square is $(335.2 / 941.6) = 0.356$. Thus, the reported value must be the adjusted R^2 . To verify this we see that the adjusted R-squared is: $1 - ((26 - 1) / (26 - 4 - 1)) \times (1 - 0.356) = 0.233$. Note that whenever there is more than one independent variable, the adjusted R^2 will always be less than R^2 .

(Module 1.2, LOS 1.e)

Question #16 - 16 of 164

Question ID: 1471975

In preparing and using this model, Williams has *least* likely relied on which of the following assumptions?

A) The residuals are homoscedastic.



B) There is a linear relationship between the independent variables.



C) The disturbance or error term is normally distributed.



Explanation

Multiple regression models assume that there is no linear relationship between two or more of the independent variables. The other answer choices are both assumptions of multiple regression.

(Module 1.2, LOS 1.e)

Question #17 of 164

Question ID: 1479919

Which of the following is *least likely* a method of detecting serial correlations?

A) A scatter plot of the residuals over time.



B) The Breusch-Godfrey test.



C) The Breusch-Pagan test.



Explanation

The Breusch-Pagan test is a test of the heteroskedasticity and not of serial correlation.

(Module 1.3, LOS 1.i)

Question #18 of 164

Question ID: 1471873

Henry Hilton, CFA, is undertaking an analysis of the bicycle industry. He hypothesizes that bicycle sales (SALES) are a function of three factors: the population under 20 (POP), the level of disposable income (INCOME), and the number of dollars spent on advertising (ADV). All data are measured in millions of units. Hilton gathers data for the last 20 years and estimates the following equation (standard errors in parentheses):

SALES	= 0.000	+ 0.004 POP	+ 1.031 INCOME	+ 2.002 ADV
	(0.113)	(0.005)	(0.337)	(2.312)

For next year, Hilton estimates the following parameters: (1) the population under 20 will be 120 million, (2) disposable income will be \$300,000,000, and (3) advertising expenditures will be \$100,000,000. Based on these estimates and the regression equation, what are predicted sales for the industry for next year?

A) \$509,980,000.



B) \$557,143,000.



C) \$656,991,000.

**Explanation**

Predicted sales for next year are:

$$\text{SALES} = \alpha + 0.004 (120) + 1.031 (300) + 2.002 (100) = 509,980,000.$$

(Module 1.1, LOS 1.b)

Damon Washburn, CFA, is currently enrolled as a part-time graduate student at State University. One of his recent assignments for his course on Quantitative Analysis is to perform a regression analysis utilizing the concepts covered during the semester. He must interpret the results of the regression as well as the test statistics. Washburn is confident in his ability to calculate the statistics because the class is allowed to use statistical software. However, he realizes that the interpretation of the statistics will be the true test of his knowledge of regression analysis. His professor has given to the students a list of questions that must be answered by the results of the analysis.

Washburn has estimated a regression equation in which 160 quarterly returns on the S&P 500 are explained by three macroeconomic variables: employment growth (EMP) as measured by nonfarm payrolls, gross domestic product (GDP) growth, and private investment (INV). The results of the regression analysis are as follows:

Coefficient Estimates		
Parameter	Coefficient	Standard Error of Coefficient
Intercept	9.50	3.40
EMP	-4.50	1.25
GDP	4.20	0.76
INV	-0.30	0.16

Other Data:

- Regression sum of squares (RSS) = 126.00
- Sum of squared errors (SSE) = 267.00
- Durbin-Watson statistic (DW) = 1.34

Abbreviated Table of the Student's t-distribution (One-Tailed Probabilities)					
df	p = 0.10	p = 0.05	p = 0.025	p = 0.01	p = 0.005
3	1.638	2.353	3.182	4.541	5.841
10	1.372	1.812	2.228	2.764	3.169
50	1.299	1.676	2.009	2.403	2.678
100	1.290	1.660	1.984	2.364	2.626
120	1.289	1.658	1.980	2.358	2.617
200	1.286	1.653	1.972	2.345	2.601

Critical Values for Durbin-Watson Statistic ($\alpha = 0.05$)										
	K=1		K=2		K=3		K=4		K=5	
n	dl	du	dl	du	dl	du	dl	du	dl	du
20	1.20	1.41	1.10	1.54	1.00	1.68	0.90	1.83	0.79	1.99
50	1.50	1.59	1.46	1.63	1.42	1.67	1.38	1.72	1.34	1.77
>100	1.65	1.69	1.63	1.72	1.61	1.74	1.59	1.76	1.57	1.78

How many of the three independent variables (not including the intercept term) are statistically significant in explaining quarterly stock returns at the 5.0% level?

A) Two of the three are statistically significant.



B) One of the three is statistically significant.



C) All three are statistically significant.



Explanation

To determine whether the independent variables are statistically significant, we use the student's t-statistic, where t equals the coefficient estimate divided by the standard error of the coefficient. This is a two-tailed test. The critical value for a 5.0% significance level and 156 degrees of freedom ($160-3-1$) is about 1.980, according to the table.

The t-statistic for employment growth = $-4.50/1.25 = -3.60$.

The t-statistic for GDP growth = $4.20/0.76 = 5.53$.

The t-statistic for investment growth = $-0.30/0.16 = -1.88$.

Therefore, employment growth and GDP growth are statistically significant because the absolute values of their t-statistics are larger than the critical value, which means two of the three independent variables are statistically significantly different from zero.

(Module 1.1, LOS 1.b)

Question #20 - 24 of 164

Question ID: 1471894

Can the null hypothesis that the GDP growth coefficient is equal to 3.50 be rejected at the 1.0% confidence level versus the alternative that it is not equal to 3.50? The null hypothesis is:

A) not rejected because the t-statistic is equal to 0.92.



B) rejected because the t-statistic is less than 2.617.



C) accepted because the t-statistic is less than 2.617.



Explanation

The hypothesis is:

$$H_0: b_{GDP} = 3.50$$

$$H_a: b_{GDP} \neq 3.50$$

This is a two-tailed test. The critical value for the 1.0% significance level and 156 degrees of freedom ($160 - 3 - 1$) is about 2.617. The t-statistic is $(4.20 - 3.50)/0.76 = 0.92$. Because the t-statistic is less than the critical value, we cannot reject the null hypothesis. Notice we cannot say that the null hypothesis is accepted; only that it is not rejected.

(Module 1.1, LOS 1.b)

Question #21 - 24 of 164

Question ID: 1471895

The percentage of the total variation in quarterly stock returns explained by the independent variables is *closest* to:

A) 32%.



B) 47%.



C) 42%.



Explanation

The R^2 is the percentage of variation in the dependent variable explained by the independent variables. The R^2 is equal to the $SS_{\text{Regression}}/SS_{\text{Total}}$, where the SS_{Total} is equal to $SS_{\text{Regression}} + SS_{\text{Error}}$. $R^2 = 126.00 / (126.00 + 267.00) = 32\%$.

(Module 1.1, LOS 1.b)

Question #22 - 24 of 164

Question ID: 1471896

According to the Durbin-Watson statistic, there is:

A) significant heteroskedasticity in the residuals.



B) no significant positive serial correlation in the residuals.



C) significant positive serial correlation in the residuals.



Explanation

The Durbin-Watson statistic tests for serial correlation in the residuals. According to the table, $d_l = 1.61$ and $d_u = 1.74$ for three independent variables and 160 degrees of freedom. Because the DW (1.34) is less than the lower value (1.61), the null hypothesis of no significant positive serial correlation can be rejected. This means there is a problem with serial correlation in the regression, which affects the interpretation of the results.

(Module 1.1, LOS 1.b)

Question #23 - 24 of 164

Question ID: 1471897

What is the predicted quarterly stock return, given the following forecasts?

- Employment growth = 2.0%
- GDP growth = 1.0%
- Private investment growth = -1.0%

A) 4.7%.



B) 5.0%.



C) 4.4%.



Explanation

Predicted quarterly stock return is $9.50\% + (-4.50)(2.0\%) + (4.20)(1.0\%) + (-0.30)(-1.0\%) = 5.0\%$.

(Module 1.1, LOS 1.b)

Question #24 - 24 of 164

Question ID: 1471898

What is the standard error of the estimate?

A) 1.71.



B) 1.31.



C) 0.81.



Explanation

The standard error of the estimate is equal to $[SSE/(n - k - 1)]^{1/2} = [267.00/156]^{1/2} =$ approximately 1.31.

(Module 1.1, LOS 1.b)

Question #25 of 164

Question ID: 1472067

An analyst runs a regression of portfolio returns on three independent variables. These independent variables are price-to-sales (P/S), price-to-cash flow (P/CF), and price-to-book (P/B). The analyst discovers that the p-values for each independent variable are relatively high. However, the F-test has a very small p-value. The analyst is puzzled and tries to figure out how the F-test can be statistically significant when the individual independent variables are not significant. What violation of regression analysis has occurred?

A) serial correlation.



B) conditional heteroskedasticity.



C) multicollinearity.



Explanation

An indication of multicollinearity is when the independent variables individually are not statistically significant but the F-test suggests that the variables as a whole do an excellent job of explaining the variation in the dependent variable.

(Module 1.3, LOS 1.j)

Question #26 of 164

Question ID: 1479904

Which of the following is *least likely* to result in misspecification of a regression model?

A) Omission of an important independent variable.



B) Inappropriate variable form.



C) Transforming a variable.



Explanation

The four types of model specification errors are: omission of an important independent variable, inappropriate variable form, inappropriate variable scaling and data improperly pooled. Transforming an independent variable is usually done to rectify inappropriate variable scaling.

(Module 1.3, LOS 1.g)

Question #27 of 164

Question ID: 1471947

An analyst regresses the return of a S&P 500 index fund against the S&P 500, and also regresses the return of an active manager against the S&P 500. The analyst uses the last five years of data in both regressions. Without making any other assumptions, which of the following is *most* accurate? The index fund:

- A) should have a higher coefficient on the independent variable. 
- B) regression should have higher sum of squares regression as a ratio to the total sum of squares. 
- C) should have a lower coefficient of determination. 

Explanation

The index fund regression should provide a higher R^2 than the active manager regression. R^2 is the sum of squares regression divided by the total sum of squares.

(Module 1.2, LOS 1.d)

Question #28 of 164

Question ID: 1471867

Consider the following estimated regression equation, with calculated t -statistics of the estimates as indicated:

$$\text{AUTO}_t = 10.0 + 1.25 \text{PI}_t + 1.0 \text{TEEN}_t - 2.0 \text{INS}_t$$

with a PI calculated t -statistic of 0.45, a TEEN calculated t -statistic of 2.2, and an INS calculated t -statistic of 0.63.

The equation was estimated over 40 companies. Using a 5% level of significance, which of the independent variables significantly different from zero?

- A) PI and INS only. 
- B) TEEN only. 
- C) PI only. 

Explanation

The critical t -values for $40-3-1 = 36$ degrees of freedom and a 5% level of significance are ± 2.028 . Therefore, only TEEN is statistically significant.

(Module 1.1, LOS 1.b)

Question #29 of 164

Question ID: 1472066

When two or more of the independent variables in a multiple regression are correlated with each other, the condition is called:

- A) serial correlation. 
- B) multicollinearity. 
- C) conditional heteroskedasticity. 

Explanation

Multicollinearity refers to the condition when two or more of the independent variables, or linear combinations of the independent variables, in a multiple regression are highly correlated with each other. This condition distorts the standard error of estimate and the coefficient standard errors, leading to problems when conducting *t*-tests for statistical significance of parameters.

(Module 1.3, LOS 1.j)

Question #30 of 164

Question ID: 1471868

Consider the following estimated regression equation, with the standard errors of the slope coefficients as noted:

$$\text{Sales}_i = 10.0 + 1.25 \text{ R\&D}_i + 1.0 \text{ ADV}_i - 2.0 \text{ COMP}_i + 8.0 \text{ CAP}_i$$

where the standard error for the estimated coefficient on R&D is 0.45, the standard error for the estimated coefficient on ADV is 2.2, the standard error for the estimated coefficient on COMP is 0.63, and the standard error for the estimated coefficient on CAP is 2.5.

The equation was estimated over 40 companies. Using a 5% level of significance, which of the estimated coefficients are significantly different from zero?

- A) R&D, ADV, COMP, and CAP. 
- B) R&D, COMP, and CAP only. 
- C) ADV and CAP only. 

Explanation

The critical t-values for $40-4-1 = 35$ degrees of freedom and a 5% level of significance are ± 2.03 .

The calculated t-values are:

$$t \text{ for R\&D} = 1.25 / 0.45 = 2.777$$

$$t \text{ for ADV} = 1.0 / 2.2 = 0.455$$

$$t \text{ for COMP} = -2.0 / 0.63 = -3.175$$

$$t \text{ for CAP} = 8.0 / 2.5 = 3.2$$

Therefore, R&D, COMP, and CAP are statistically significant.

(Module 1.1, LOS 1.b)

Consider a study of 100 university endowment funds that was conducted to determine if the funds' annual risk-adjusted returns could be explained by the size of the fund and the percentage of fund assets that are managed to an indexing strategy. The equation used to model this relationship is:

$$ARAR_i = b_0 + b_1 \text{Size}_i + b_2 \text{Index}_i + e_i$$

Where:

$ARAR_i$ = the average annual risk-adjusted percent returns for the fund i over the 1998-2002 time period.

Size_i = the natural logarithm of the average assets under management for fund i .

Index_i = the percentage of assets in fund i that were managed to an indexing strategy.

The table below contains a portion of the regression results from the study.

Partial Results from Regression ARAR on Size and Extent of Indexing			
	Coefficients	Standard Error	t-Statistic
Intercept	???	0.55	-5.2
Size	0.6	0.18	???
Index	1.1	???	2.1

Which of the following is the *most* accurate interpretation of the slope coefficient for size?

ARAR:

- A) will change by 0.6% when the natural logarithm of assets under management changes by 1.0, holding index constant. 
- B) will change by 1.0% when the natural logarithm of assets under management changes by 0.6, holding index constant. 
- C) and index will change by 1.1% when the natural logarithm of assets under management changes by 1.0. 

Explanation

A slope coefficient in a multiple linear regression model measures how much the dependent variable changes for a one-unit change in the independent variable, *holding all other independent variables constant*. In this case, the independent variable size (= ln average assets under management) has a slope coefficient of 0.6, indicating that the dependent variable ARAR will change by 0.6% return for a one-unit change in size, assuming nothing else changes. Pay attention to the units on the dependent variable.

(Module 1.1, LOS 1.b)

Question #32 - 36 of 164

Question ID: 1471876

Which of the following is the estimated standard error of the regression coefficient for index?

- A) 2.31. 
- B) 1.91. 
- C) 0.52. 

Explanation

The t -statistic for testing the null hypothesis $H_0: \beta_i = 0$ is $t = (b_i - 0) / \beta_i$, where β_i is the population parameter for independent variable i , b_i is the estimated coefficient, and β_i is the coefficient standard error. Using the information provided, the estimated coefficient standard error can be computed as $b_{\text{Index}} / t = \beta_{\text{Index}} = 1.1 / 2.1 = 0.5238$.

(Module 1.1, LOS 1.b)

Question #33 - 36 of 164

Question ID: 1471877

Which of the following is the t -statistic for size?

- A) 0.30.
- B) 3.33.
- C) 0.70.



Explanation

The t -statistic for testing the null hypothesis $H_0: \beta_i = 0$ is $t = (b_i - 0) / \sigma_i$, where β_i is the population parameter for independent variable i , b_i is the estimated coefficient, and σ_i is the coefficient standard error. Using the information provided, the t -statistic for size can be computed as $t = b_{\text{Size}} / \sigma_{\text{Size}} = 0.6 / 0.18 = 3.3333$.

(Module 1.1, LOS 1.b)

Question #34 - 36 of 164

Question ID: 1471878

Which of the following is the estimated intercept for the regression?

- A) -2.86.
- B) -9.45.
- C) -0.11.



Explanation

The t -statistic for testing the null hypothesis $H_0: \beta_i = 0$ is $t = (b_i - 0) / \sigma_i$, where σ_i is the population parameter for independent variable i , b_i is the estimated parameter, and σ_i is the parameter's standard error. Using the information provided, the estimated intercept can be computed as $b_0 = t \times \sigma_0 = -5.2 \times 0.55 = -2.86$.

(Module 1.1, LOS 1.b)

Question #35 - 36 of 164

Question ID: 1471879

Which of the following statements is *most* accurate regarding the significance of the regression parameters at a 5% level of significance?

- A) The parameter estimates for the intercept are significantly different than zero.
The slope coefficients for index and size are not significant.
- B) All of the parameter estimates are significantly different than zero at the 5% level of significance.



- C) The parameter estimates for the intercept and the independent variable size are significantly different than zero. The coefficient for index is not significant.



Explanation

At 5% significance and 97 degrees of freedom ($100 - 3$), the critical t -value is slightly greater than, but very close to, 1.984. The t -statistic for the intercept and *index* are provided as -5.2 and 2.1 , respectively, and the t -statistic for size is computed as $0.6 / 0.18 = 3.33$. The absolute value of all of the regression intercepts is greater than $t_{\text{critical}} = 1.984$. Thus, it can be concluded that all of the parameter estimates are significantly different than zero at the 5% level of significance.

(Module 1.1, LOS 1.b)

Question #36 - 36 of 164

Question ID: 1471880

Which of the following is NOT a required assumption for multiple linear regression?

- A) The expected value of the error term is zero.
- B) The error term is linearly related to the dependent variable.
- C) The error term is normally distributed.



Explanation

The assumptions of multiple linear regression include: linear relationship between dependent and independent variable, independent variables are not random and no exact linear relationship exists between the two or more independent variables, error term is normally distributed with an expected value of zero and constant variance, and the error term is serially uncorrelated.

(Module 1.1, LOS 1.b)

Question #37 of 164

Question ID: 1471980

An analyst runs a regression of monthly value-stock returns on five independent variables over 48 months. The total sum of squares is 430, and the sum of squared errors is 170. Test the null hypothesis at the 2.5% and 5% significance level that all five of the independent variables are equal to zero.

- A) Rejected at 5% significance only.
- B) Not rejected at 2.5% or 5.0% significance.
- C) Rejected at 2.5% significance and 5% significance.



Explanation

The F-statistic is equal to the ratio of the mean squared regression (MSR) to the mean squared error (MSE).

$$RSS = SST - SSE = 430 - 170 = 260$$

$$MSR = 260 / 5 = 52$$

$$MSE = 170 / (48 - 5 - 1) = 4.05$$

$$F = 52 / 4.05 = 12.84$$

The critical F-value for 5 and 42 degrees of freedom at a 5% significance level is approximately 2.44. The critical F-value for 5 and 42 degrees of freedom at a 2.5% significance level is approximately 2.89. Therefore, we can reject the null hypothesis at either level of significance and conclude that at least one of the five independent variables explains a significant portion of the variation of the dependent variable.

(Module 1.2, LOS 1.e)

Question #38 of 164

Question ID: 1471946

Which of the following statements regarding the R^2 is *least* accurate?

- A) It is possible for the adjusted- R^2 to decline as more variables are added to the multiple regression. 
- B) The adjusted- R^2 is greater than the R^2 in multiple regression. 
- C) The adjusted- R^2 not appropriate to use in simple regression. 

Explanation

The adjusted- R^2 will always be less than R^2 in multiple regression.

(Module 1.2, LOS 1.d)

Question #39 of 164

Question ID: 1471883

Jacob Warner, CFA, is evaluating a regression analysis recently published in a trade journal that hypothesizes that the annual performance of the S&P 500 stock index can be explained by movements in the Federal Funds rate and the U.S. Producer Price Index (PPI). Which of the following statements regarding his analysis is *most accurate*?

- A) If the p -value of a variable is less than the significance level, the null hypothesis cannot be rejected. 
- B) If the t -value of a variable is less than the significance level, the null hypothesis cannot be rejected. 
- C) If the p -value of a variable is less than the significance level, the null hypothesis can be rejected. 

Explanation

The p -value is the smallest level of significance for which the null hypothesis can be rejected. Therefore, for any given variable, if the p -value of a variable is less than the significance level, the null hypothesis can be rejected and the variable is considered to be statistically significant.

(Module 1.1, LOS 1.b)

Autumn Voiku is attempting to forecast sales for Brookfield Farms based on a multiple regression model. Voiku has constructed the following model:

$$\text{sales} = b_0 + (b_1 \times \text{CPI}) + (b_2 \times \text{IP}) + (b_3 \times \text{GDP}) + \varepsilon_t$$

Where:

sales = \$ change in sales (in 000's)

CPI = change in the consumer price index

IP = change in industrial production (millions)

GDP = change in GDP (millions)

All changes in variables are in percentage terms.

Voiku uses monthly data from the previous 180 months of sales data and for the independent variables. The model estimates (with coefficient standard errors in parentheses) are:

SALES =	10.2	+ (4.6 × CPI)	+ (5.2 × IP)	+ (11.7 × GDP)
	(5.4)	(3.5)	(5.9)	(6.8)

The sum of squared errors is 140.3 and the total sum of squares is 368.7.

Voiku calculates the unadjusted R^2 , the adjusted R^2 , and the standard error of estimate to be 0.592, 0.597, and 0.910, respectively.

Voiku is concerned that one or more of the assumptions underlying multiple regression has been violated in her analysis. In a conversation with Dave Grimble, CFA, a colleague who is considered by many in the firm to be a quant specialist, Voiku says, "It is my understanding that there are five assumptions of a multiple regression model:"

Assumption 1:	There is a linear relationship between the dependent and independent variables.
Assumption 2:	The independent variables are not random, and there is zero correlation between any two of the independent variables.
Assumption 3:	The residual term is normally distributed with an expected value of zero.
Assumption 4:	The residuals are serially correlated.
Assumption 5:	The variance of the residuals is constant.

Grimble agrees with Miller's assessment of the assumptions of multiple regression.

Voiku tests and fails to reject each of the following four null hypotheses at the 99% confidence interval:

Hypothesis 1:	The coefficient on GDP is negative.
Hypothesis 2:	The intercept term is equal to -4.
Hypothesis 3:	A 2.6% increase in the CPI will result in an increase in sales of more than 12.0%.
Hypothesis 4:	A 1% increase in industrial production will result in a 1% decrease in sales.

Figure 1: Partial table of the Student's *t*-distribution (One-tailed probabilities)

df	p = 0.10	p = 0.05	p = 0.025	p = 0.01	p = 0.005
170	1.287	1.654	1.974	2.348	2.605
176	1.286	1.654	1.974	2.348	2.604
180	1.286	1.653	1.973	2.347	2.603

Figure 2: Partial *F*-Table critical values for right-hand tail area equal to 0.05

	df1 = 1	df1 = 3	df1 = 5
df2 = 170	3.90	2.66	2.27

$df2 = 176$	3.89	2.66	2.27
$df2 = 180$	3.89	2.65	2.26

Figure 3: Partial F-Table critical values for right-hand tail area equal to 0.025

	df1 = 1	df1 = 3	df1 = 5
$df2 = 170$	5.11	3.19	2.64
$df2 = 176$	5.11	3.19	2.64
$df2 = 180$	5.11	3.19	2.64

Question #40 - 45 of 164

Question ID: 1471907

Concerning the assumptions of multiple regression, Grimbles is:

- A) incorrect to agree with Voiku's list of assumptions because two of the assumptions are stated incorrectly. ✓
- B) incorrect to agree with Voiku's list of assumptions because one of the assumptions is stated incorrectly. ✗
- C) correct to agree with Voiku's list of assumptions. ✗

Explanation

Assumption 2 is stated incorrectly. Some correlation between independent variables is unavoidable; and high correlation results in multicollinearity. However, an exact linear relationship between linear combinations of two or more independent variables should not exist.

Assumption 4 is also stated incorrectly. The assumption is that the residuals are serially uncorrelated (i.e., they are not serially correlated).

(Module 1.1, LOS 1.b)

Question #41 - 45 of 164

Question ID: 1471908

For which of the four hypotheses did Voiku incorrectly fail to reject the null, based on the data given in the problem?

- A) Hypothesis 2. ✓
- B) Hypothesis 4. ✗

C) Hypothesis 3.



Explanation

The critical values at the 1% level of significance (99% confidence) are 2.348 for a one-tail test and 2.604 for a two-tail test ($df = 176$).

The t-values for the hypotheses are:

Hypothesis 1: $11.7 / 6.8 = 1.72$

Hypothesis 2: $14.2 / 5.4 = 2.63$

Hypothesis 3: $12.0 / 2.6 = 4.6$, so the hypothesis is that the coefficient is greater than 4.6, and the t-stat of that hypothesis is $(4.6 - 4.6) / 3.5 = 0$.

Hypothesis 4: $(5.2 + 1) / 5.9 = 1.05$

Hypotheses 1 and 3 are one-tail tests; 2 and 4 are two-tail tests. Only Hypothesis 2 exceeds the critical value, so only Hypothesis 2 should be rejected.

(Module 1.1, LOS 1.b)

Question #42 - 45 of 164

Question ID: 1471909

The *most* appropriate decision with regard to the F-statistic for testing the null hypothesis that all of the independent variables are simultaneously equal to zero at the 5 percent significance level is to:

- A) reject the null hypothesis because the F-statistic is larger than the critical F-value of 3.19. 
- B) fail to reject the null hypothesis because the F-statistic is smaller than the critical F-value of 2.66. 
- C) reject the null hypothesis because the F-statistic is larger than the critical F-value of 2.66. 

Explanation

$RSS = 368.7 - 140.3 = 228.4$, $F\text{-statistic} = (228.4 / 3) / (140.3 / 176) = 95.51$. The critical value for a one-tailed 5% F-test with 3 and 176 degrees of freedom is 2.66. Because the F-statistic is greater than the critical F-value, the null hypothesis that all of the independent variables are simultaneously equal to zero should be rejected.

(Module 1.1, LOS 1.b)

Question #43 - 45 of 164

Question ID: 1471910

Regarding Voiku's calculations of R^2 and the standard error of estimate, she is:

- A) correct in her calculation of the unadjusted R^2 but incorrect in her calculation of the standard error of estimate. 
- B) incorrect in her calculation of the unadjusted R^2 but correct in her calculation of the standard error of estimate. 
- C) incorrect in her calculation of both the unadjusted R^2 and the standard error of estimate. 

Explanation

$$SEE = \sqrt{[140.3 / (180 - 3 - 1)]} = 0.893$$

$$\text{unadjusted } R^2 = (368.7 - 140.3) / 368.7 = 0.619$$

(Module 1.1, LOS 1.b)

Question #44 - 45 of 164

Question ID: 1471911

The multiple regression, as specified, *most likely* suffers from:

- A) heteroskedasticity. 
- B) serial correlation of the error terms. 
- C) multicollinearity. 

Explanation

The regression is highly significant (based on the F-stat in Part 3), but the individual coefficients are not. This is a result of a regression with significant multicollinearity problems. The t-stats for the significance of the regression coefficients are, respectively, 1.89, 1.31, 0.88, 1.72. None of these are high enough to reject the hypothesis that the coefficient is zero at the 5% level of significance (two-tailed critical value of 1.974 from t-table).

(Module 1.1, LOS 1.b)

Question #45 - 45 of 164

Question ID: 1471912

A 90 percent confidence interval for the coefficient on GDP is:

- A) -1.5 to 20.0. 

B) -1.9 to 19.6.



C) 0.5 to 22.9.



Explanation

A 90% confidence interval with 176 degrees of freedom is coefficient $\pm t_c(s_e) = 11.7 \pm 1.654$ (6.8) or 0.5 to 22.9.

(Module 1.1, LOS 1.b)

In preparing an analysis of HB Inc., Jack Stumper is asked to look at the company's sales in relation to broad based economic indicators. Stumper's analysis indicates that HB's monthly sales are related to changes in housing starts (H) and changes in the mortgage interest rate (M). The analysis covers the past ten years for these variables. The regression equation is:

$$S = 1.76 + 0.23H - 0.08M$$

Number of observations:	123
Unadjusted R ² :	0.77
F statistic:	9.80
Durbin Watson statistic	0.50
p-value of Housing Starts	0.017
t-stat of Mortgage Rates	-2.6

Variable Descriptions

S = HB Sales (in thousands)

H = housing starts (in thousands)

M = mortgage interest rate (in percent)

November 20x6 Actual Data

HB's monthly sales: \$55,000

Housing starts: 150,000

Mortgage interest rate (%): 7.5

Critical Values for Student's t-Distribution

Degrees of Freedom	Level of significance for one-tailed test					

	10%	5%	2.5%	1%	0.5%	0.05%
	Level of significance for two-tailed test					
	20%	10%	5%	2%	1%	0.1%
10	1.372	1.812	2.228	2.764	3.169	4.587
20	1.325	1.725	2.086	2.528	2.845	3.850
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
120	1.289	1.658	1.980	2.358	2.617	3.373

Question #46 - 51 of 164

Question ID: 1471914

Using the regression model developed, the closest prediction of sales for December 20x6 is:

A) \$55,000.



B) \$36,000.



C) \$44,000.



Explanation

$$1.76 + 0.23 * (150) - 0.08 * (7.5) = 35.66.$$

(Module 1.1, LOS 1.b)

Question #47 - 51 of 164

Question ID: 1471915

Will Stumper conclude that the housing starts coefficient is statistically different from zero and how will he interpret it at the 5% significance level:

A) different from zero; sales will rise by \$23 for every 100 house starts.



B) not different from zero; sales will rise by \$0 for every 100 house starts.



C) different from zero; sales will rise by \$100 for every 23 house starts.



Explanation

A p-value (0.017) below significance (0.05) indicates a variable which is statistically different from zero. The coefficient of 0.23 indicates that sales will rise by \$23 for every 100 house starts.

(Module 1.1, LOS 1.b)

Question #48 - 51 of 164

Question ID: 1471916

Is the regression coefficient of changes in mortgage interest rates different from zero at the 5 percent level of significance?

- A) no, because $2.6 < 2.62$.
- B) yes, because $2.6 > 2.23$.
- C) yes, because $2.6 > 1.98$.

**Explanation**

The correct degrees of freedom for critical t-statistic is $n-k-1 = 123-2-1 = 120$. From the t-table, 5% L.O.S., 2-tailed, critical t-value is 1.98. Note that the t-stat for the coefficient for mortgage rate is directly given in the question (-2.6).

(Module 1.1, LOS 1.b)

Question #49 - 51 of 164

Question ID: 1471917

In this multiple regression, the F-statistic indicates the:

- A) degree of correlation between the independent variables.
- B) deviation of the estimated values from the actual values of the dependent variable.
- C) the joint significance of the independent variables.

**Explanation**

The F-statistic indicates the joint significance of the independent variables. The deviation of the estimated values from the actual values of the dependent variable is the standard error of estimate. The degree of correlation between the independent variables is the coefficient of correlation.

(Module 1.1, LOS 1.b)

Question #50 - 51 of 164

Question ID: 1471918

The regression statistics above indicate that for the period under study, the independent variables (housing starts, mortgage interest rate) together explained approximately what percentage of the variation in the dependent variable (sales)?

A) 77.00.



B) 9.80.



C) 67.00.



Explanation

The question is asking for the coefficient of determination.

(Module 1.1, LOS 1.b)

Question #51 - 51 of 164

Question ID: 1471919

In this multiple regression, if Stumper discovers that the residuals exhibit positive serial correlation, the *most likely* effect is:

A) standard errors are too high but coefficient estimate is consistent.



B) standard errors are too low but coefficient estimate is consistent.



C) standard errors are not affected but coefficient estimate is inconsistent.



Explanation

Positive serial correlation does not affect the consistency of coefficients (i.e., the coefficients are still consistent) but the estimated standard errors are too low leading to artificially high t-statistics.

(Module 1.1, LOS 1.b)

Ben Sasse is a quantitative analyst at Gurnop Asset Managers. Sasse is interviewing Victor Sophie for a junior analyst position. Sasse mentions that the firm currently uses several proprietary multiple regression models and wants Sophie's opinion about regression models.

Sophie makes the following statements:

Statement 1: Multiple regression models can be used to forecast independent variables.

Statement 2: Multiple regression models can be used to test existing theories of relationships among variables.

Sasse then discusses a model that the firm uses to forecast credit spread on investment-grade corporate bonds. Sasse states that while the current model parameters are a secret, the following is an older version of the model:

$$\text{CSP} = 0.22 + 1.04 \times \text{DSC} - 0.32 \times \text{index} + 1.33 \times \text{D/E}$$

where:

CSP = credit spread (%)

DSC = EBITDA / unsecured debt

index = 1 if the issuer is part of CDX index; 0 otherwise

D/E = long-term debt / equity

Question #52 - 55 of 164

Question ID: 1484384

Regarding Sophie's statement on multiple regression:

A) both statements are correct.



B) only Statement 2 is correct.



C) only Statement 1 is correct.



Explanation

Multiple regression models can be used to *identify* relations between variables, forecast the *dependent* variable, and *test* existing theories. Statement 1 is inaccurate in because it mentions forecast independent (and not dependent) variables.

(Module 1.1, LOS 1.a)

Question #53 - 55 of 164

Question ID: 1479870

Based on the credit spread model, if an issuer gets included in the CDX index and assuming everything else the same, which of the following statements *most accurately* describes the model's forecast?

A) The credit spread on the firm's issue will increase by 32 bps.



B) The credit spread on the firm's issue will decrease by 32 bps.



C) The credit spread on the firm's issue would decrease by 10 bps.



Explanation

The coefficient on the index dummy variable is -0.32 , and if the variable takes a value of 1 (inclusion in the index), the credit spread would decrease by 0.32%, or 32 bps.

(Module 1.1, LOS 1.b)

Question #54 - 55 of 164

Question ID: 1479871

Which of the following is *least likely* an assumption of multiple linear regression?

- A) There is no linear relationship between the independent variables. 
- B) The error term is normally distributed. 
- C) The dependent variable is not serially correlated. 

Explanation

The assumption calls for the residual (or errors) to be not serially correlated. The dependent variable can have serial correlation. Other assumptions are accurate.

(Module 1.1, LOS 1.c)

Question #55 - 55 of 164

Question ID: 1479872

Which assumption of multiple regression is *most likely* evaluated using a QQ plot?

- A) Error term is normally distributed. 
- B) Conditional heteroskedasticity. 
- C) Serial correlation of residuals. 

Explanation

A normal QQ plot of the residuals can visually indicate violation of the assumption that the residuals are normally distributed.

(Module 1.1, LOS 1.c)

Miles Mason, CFA, works for ABC Capital, a large money management company based in New York. Mason has several years of experience as a financial analyst, but is currently working in the marketing department developing materials to be used by ABC's sales team for both existing and prospective clients. ABC Capital's client base consists primarily of large net worth individuals and Fortune 500 companies. ABC invests its clients' money in both

publicly traded mutual funds as well as its own investment funds that are managed in-house. Five years ago, roughly half of its assets under management were invested in the publicly traded mutual funds, with the remaining half in the funds managed by ABC's investment team. Currently, approximately 75% of ABC's assets under management are invested in publicly traded funds, with the remaining 25% being distributed among ABC's private funds. The managing partners at ABC would like to shift more of its client's assets away from publicly traded funds into ABC's proprietary funds, ultimately returning to a 50/50 split of assets between publicly traded funds and ABC funds. There are three key reasons for this shift in the firm's asset base. First, ABC's in-house funds have outperformed other funds consistently for the past five years. Second, ABC can offer its clients a reduced fee structure on funds managed in-house relative to other publicly traded funds. Lastly, ABC has recently hired a top fund manager away from a competing investment company and would like to increase his assets under management.

ABC Capital's upper management requested that current clients be surveyed in order to determine the cause of the shift of assets away from ABC funds. Results of the survey indicated that clients feel there is a lack of information regarding ABC's funds. Clients would like to see extensive information about ABC's past performance, as well as a sensitivity analysis showing how the funds will perform in varying market scenarios. Mason is part of a team that has been charged by upper management to create a marketing program to present to both current and potential clients of ABC. He needs to be able to demonstrate a history of strong performance for the ABC funds, and, while not promising any measure of future performance, project possible return scenarios. He decides to conduct a regression analysis on all of ABC's in-house funds. He is going to use 12 independent economic variables in order to predict each particular fund's return. Mason is very aware of the many factors that could minimize the effectiveness of his regression model, and if any are present, he knows he must determine if any corrective actions are necessary. Mason is using a sample size of 121 monthly returns.

Question #56 - 58 of 164

Question ID: 1480342

Which of the following tests is *least likely* to be used to detect autocorrelation?

- A) Durbin-Watson.
- B) Breusch-Godfrey
- C) Breusch-Pagan.



Explanation

Durbin-Watson and Breusch-Godfrey test statistic are used to detect autocorrelation. The Breusch-Pagan test is used to detect heteroskedasticity.

(Module 1.3, LOS 1.i)

Question #57 - 58 of 164

Question ID: 1472023

One of the most popular ways to correct heteroskedasticity is to:

- A) improve the specification of the model.
- B) adjust the standard errors.
- C) use robust standard errors.



Explanation

Using generalized least squares and calculating robust standard errors are possible remedies for heteroskedasticity. Improving specifications remedies serial correlation. The standard error cannot be adjusted, only the coefficient of the standard errors.

(Module 1.3, LOS 1.h)

Question #58 - 58 of 164

Question ID: 1479933

If a regression equation shows that no individual t-tests are significant, but the F-statistic is significant, the regression probably exhibits:

- A) serial correlation.
- B) multicollinearity.
- C) heteroskedasticity.



Explanation

Common indicators of multicollinearity include: high correlation (>0.7) between independent variables, no individual t-tests are significant but the F-statistic is, and signs on the coefficients that are opposite of what is expected.

(Module 1.3, LOS 1.j)

George Smith, an analyst with Great Lakes Investments, has created a comprehensive report on the pharmaceutical industry at the request of his boss. The Great Lakes portfolio currently has a significant exposure to the pharmaceuticals industry through its large equity position in the top two pharmaceutical manufacturers. His boss requested that Smith determine a way to accurately forecast pharmaceutical sales in order for Great Lakes to identify further investment opportunities in the industry as well as to minimize their exposure to downturns in the market. Smith realized that there are many factors that could possibly have an impact on sales, and he must identify a method that can quantify their effect. Smith used a multiple regression analysis with five independent variables to predict industry sales. His goal is to not only identify relationships that are statistically significant, but economically significant as well. The assumptions of his model are fairly standard: a linear relationship exists between the dependent and independent variables, the independent variables are not random, and the expected value of the error term is zero.

Smith is confident with the results presented in his report. He has already done some hypothesis testing for statistical significance, including calculating a t-statistic and conducting a two-tailed test where the null hypothesis is that the regression coefficient is equal to zero versus the alternative that it is not. He feels that he has done a thorough job on the report and is ready to answer any questions posed by his boss.

However, Smith's boss, John Sutter, is concerned that in his analysis, Smith has ignored several potential problems with the regression model that may affect his conclusions. He knows that when any of the basic assumptions of a regression model are violated, any results drawn for the model are questionable. He asks Smith to go back and carefully examine the effects of heteroskedasticity, multicollinearity, and serial correlation on his model. In specific, he wants Smith to make suggestions regarding how to detect these errors and to correct problems that he encounters.

Question #59 - 62 of 164

Question ID: 1479940

Sutter has detected the presence of conditional heteroskedasticity in Smith's report. This is evidence that:

- A) the error terms are correlated with each other. 
- B) the variance of the error term is correlated with the values of the independent variables. 
- C) two or more of the independent variables are highly correlated with each other. 

Explanation

Conditional heteroskedasticity exists when the variance of the error term is correlated with the values of the independent variables.

Multicollinearity, on the other hand, occurs when two or more of the independent variables are highly correlated with each other. Serial correlation exists when the error terms are correlated with each other.

(Module 1.3, LOS 1.j)

Question #60 - 62 of 164

Question ID: 1479941

Suppose there is evidence that the variance of the error term is correlated with the values of the independent variables. The *most likely* effect on the statistical inferences Smith can make from the regressions results using financial data is to commit a:

- A) Type I error by incorrectly failing to reject the null hypothesis that the regression parameters are equal to zero. 
- B) Type II error by incorrectly failing to reject the null hypothesis that the regression parameters are equal to zero. 
- C) Type I error by incorrectly rejecting the null hypotheses that the regression parameters are equal to zero. 

Explanation

One problem with conditional heteroskedasticity while working with financial data, is that the standard errors of the parameter estimates will be too small and the t-statistics too large. This will lead Smith to incorrectly reject the null hypothesis that the parameters are equal to zero. In other words, Smith will incorrectly conclude that the parameters are statistically significant when in fact they are not. This is an example of a Type I error: incorrectly rejecting the null hypothesis when it should not be rejected.

(Module 1.3, LOS 1.h)

Question #61 - 62 of 164

Question ID: 1479942

Which of the following is *most likely* to indicate that two or more of the independent variables, or linear combinations of independent variables, may be highly correlated with each other? Unless otherwise noted, significant and insignificant mean significantly different from zero and not significantly different from zero, respectively.

- A) The R^2 is low, the F-statistic is insignificant and the Durbin-Watson statistic is significant. 
- B) The R^2 is high, the F-statistic is significant and the t-statistics on the individual slope coefficients are insignificant. 
- C) The R^2 is high, the F-statistic is significant and the t-statistics on the individual slope coefficients are significant. 

Explanation

Multicollinearity occurs when two or more of the independent variables, or linear combinations of independent variables, may be highly correlated with each other. In a classic effect of multicollinearity, the R^2 is high and the F-statistic is significant, but the t-statistics on the individual slope coefficients are insignificant.

(Module 1.3, LOS 1.j)

Question #62 - 62 of 164

Question ID: 1479943

Using the Durbin-Watson test statistic, Smith rejects the null hypothesis suggested by the test. This is evidence that:

- A) two or more of the independent variables are highly correlated with each other. 
- B) the error term is normally distributed. 
- C) the error terms are correlated with each other. 

Explanation

Serial correlation (also called autocorrelation) exists when the error terms are correlated with each other.

Multicollinearity, on the other hand, occurs when two or more of the independent variables are highly correlated with each other. One assumption of multiple regression is that the error term is normally distributed.

(Module 1.3, LOS 1.i)

In preparing an analysis of Treefell Company, Jack Lumber is asked to look at the company's sales in relation to broad-based economic indicators. Lumber's analysis indicates that Treefell's monthly sales are related to changes in housing starts (H) and changes in the

mortgage interest rate (M). The analysis covers the past 10 years for these variables. The regression equation is:

$$S = 1.76 + 0.23H - 0.08M$$

Number of observations: 123

Unadjusted R^2 : 0.77

F-statistic: 9.80

Durbin-Watson statistic: 0.50

p-value of Housing Starts: 0.017

t-stat of Mortgage Rates: -2.6

Variable Descriptions

S = Treefell Sales (in thousands)

H = housing starts (in thousands)

M = mortgage interest rate (in percent)

November 20X6 Actual Data

Treefell's monthly sales: \$55,000

Housing starts: 150,000

Mortgage interest rate (%): 7.5

Partial Chi-Square Table (5% Level of Significance)

Degrees of Freedom	Critical Value
1	3.84
2	5.99

3	7.81
4	9.49
5	11.07
6	12.59

Question #63 - 67 of 164

Question ID: 1472050

Using the regression model developed, the *closest* prediction of sales for December 20X6 is:

- A) \$55,000. 
- B) \$44,000. 
- C) \$36,000. 

Explanation

$$1.76 + 0.23 \times (150) - 0.08 \times (7.5) = 35.66.$$

(Module 1.2, LOS 1.f)

Question #64 - 67 of 164

Question ID: 1472051

Will Jack conclude that the housing starts coefficient is statistically different from zero and how will he interpret it at the 5% significance level?

- A) Different from zero; sales will rise by \$100 for every 23 house starts. 
- B) Different from zero; sales will rise by \$23 for every 100 house starts. 
- C) Not different from zero; sales will rise by \$0 for every 100 house starts. 

Explanation

A p -value (0.017) below significance (0.05) indicates a variable that is statistically different from zero. The coefficient of 0.23 indicates that sales will rise by \$100 for every 23 house starts. Remember the rule $p\text{-value} < \text{significance}$, then reject null.

(Module 1.1, LOS 1.b)

Question #65 - 67 of 164

Question ID: 1479916

In this multiple regression, the F -statistic indicates the:

- A) deviation of the estimated values from the actual values of the dependent variable. 
- B) degree of correlation between the independent variables. 
- C) the joint significance of the independent variables. 

Explanation

The F -statistic is for the general linear F -test to test the null hypothesis that slope coefficients on all variables are equal to zero.

(Module 1.2, LOS 1.e)

Question #66 - 67 of 164

Question ID: 1472054

The regression statistics indicate that for the period under study, the independent variables (housing starts, mortgage interest rate) together explain approximately what percentage of the variation in the dependent variable (sales)?

- A) 9.80. 
- B) 77.00. 
- C) 67.00. 

Explanation

The question is asking for the coefficient of determination.

(Module 1.2, LOS 1.d)

Question #67 - 67 of 164

Question ID: 1484386

For this question only, assume that the regression of squared residuals on the independent variables has $R^2 = 11\%$. At a 5% level of significance, which of the following conclusions is *most* accurate?

- A) With a test statistic of 0.22, we cannot reject the null hypothesis of no conditional heteroskedasticity. 
- B) Because the critical value is 3.84, we reject the null hypothesis of no conditional heteroskedasticity. 

- C) With a test statistic of 13.53, we can conclude the presence of conditional heteroskedasticity.



Explanation

Chi-square = $n \times R^2 = 123 \times 0.11 = 13.53$. Critical Chi-square (degree of freedom = $k = 2$) = 5.99. Because the test statistic exceeds the critical value, we reject the null hypothesis (of no conditional heteroskedasticity).

(Module 1.3, LOS 1.h)

Question #68 of 164

Question ID: 1472075

Jill Wentraub is an analyst with the retail industry. She is modeling a company's sales over time and has noticed a quarterly seasonal pattern. If she includes dummy variables to represent the seasonality component of the sales she must use:

- A) four dummy variables.
- B) three dummy variables.
- C) one dummy variables.



Explanation

Three. Always use one less dummy variable than the number of possibilities. For a seasonality that varies by quarters in the year, three dummy variables are needed.

(Module 1.4, LOS 1.I)

Question #69 of 164

Question ID: 1479883

One possible problem that could jeopardize the validity of the employment growth rate model is multicollinearity. Which of the following would *most likely* suggest the existence of multicollinearity?

- A) The variance of the observations has increased over time.
- B) The Durbin–Watson statistic is significant.
- C) The F-statistic suggests that the overall regression is significant, however the regression coefficients are not individually significant.



Explanation

One symptom of multicollinearity is that the regression coefficients may not be individually statistically significant even when according to the F-statistic the overall regression is significant. The problem of multicollinearity involves the existence of high correlation between two or more independent variables. Clearly, as service employment rises, construction employment must rise to facilitate the growth in these sectors. Alternatively, as manufacturing employment rises, the service sector must grow to serve the broader manufacturing sector.

- The variance of observations suggests the possible existence of heteroskedasticity.
- If the Durbin-Watson statistic may be used to test for serial correlation at a single lag.

(Module 1.2, LOS 1.f)

Question #70 of 164

Question ID: 1479874

Wilson estimated a regression that produced the following analysis of variance (ANOVA) table:

Source	Sum of squares	Degrees of freedom	Mean square
Regression	100	1	100.0
Error	300	40	7.5
Total	400	41	

The values of R^2 and the F-statistic to test the null hypothesis that slope coefficients on all variables are equal to zero are:

A) $R^2 = 0.20$ and $F = 13.333$.



B) $R^2 = 0.25$ and $F = 0.930$.



C) $R^2 = 0.25$ and $F = 13.333$.



Explanation

$$R^2 = \text{RSS} / \text{SST} = 100 / 400 = 0.25$$

The F-statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = 100 / 7.5 = 13.333$$

(Module 1.2, LOS 1.e)

Binod Salve, CFA, is investigating the application of the Fama-French three-factor model (Model 1) for the Indian stock market for the period 2001–2011 (120 months). Using the dependent variable as annualized return (%), the results of the analysis are shown in **Indian Equities—Fama-French Model**

Indian Equities—Fama-French Model

Factor	Coefficient	P-Value	VIF
Intercept	1.22	<0.001	
SMB	0.23	<0.001	3
HML	0.34	0.003	3
Rm-Rf	0.88	<0.001	2
R-squared		0.36	
SSE		38.00	
BG (lag 1)		2.11	
BG (lag 2)		1.67	

Partial F-Table (5% Level of Significance)

Degrees of Freedom Denominator	Degrees of Freedom Numerator		
	1	2	3
112	3.93	3.08	2.69
113	3.93	3.08	2.68
114	3.92	3.08	2.68
115	3.92	3.08	2.68
116	3.92	3.07	2.68
117	3.92	3.07	2.68

Partial Chi-Square Table (5% Level of Significance)

Degrees of Freedom	Critical Value
1	3.84
2	5.99
3	7.81

4	9.49
5	11.07
6	12.59

Question #71 - 74 of 164

Question ID: 1479945

Salve runs a regression using the squared residuals from the model using the original dependent variables. The coefficient of determination of this model is 6%. Which of the following is the *most appropriate* conclusion at a 5% level of significance?

- A) Because the test statistic of 7.20 is higher than the critical value of 3.84, we reject the null hypothesis of no conditional heteroskedasticity in residuals. 
- B) Because the test statistic of 7.20 is lower than the critical value of 7.81, we reject the null hypothesis of no conditional heteroskedasticity in residuals. 
- C) Because the test statistic of 3.60 is lower than the critical value of 3.84, we reject the null hypothesis of no conditional heteroskedasticity in residuals. 

Explanation

The chi-square test statistic = $n \times R^2 = 120 \times 0.06 = 7.20$.

The one-tailed critical value for a chi-square distribution with $k = 3$ degrees of freedom and α of 5% is 7.81. Therefore, we should not reject the null hypothesis and conclude that we don't have a problem with conditional heteroskedasticity.

(Module 1.3, LOS 1.h)

Question #72 - 74 of 164

Question ID: 1479946

Which of the following misspecifications is *most likely* to cause serial correlation in residuals?

- A) Improper variable scaling. 
- B) Data improperly pooled. 
- C) Improper variable form. 

Explanation

Out of the four forms of model misspecifications, serial correlation in residuals may be caused by omission of important variables (not an answer choice) and by improper data pooling.

(Module 1.3, LOS 1.g)

Question #73 - 74 of 164

Question ID: 1480344

Should Salve be concerned about residual serial correlation?

A) Yes, for two lags only.



B) No.



C) Yes, for one lag only.



Explanation

The BG test statistic has an F -distribution with p and $n - p - k - 1$ degrees of freedom, where p = the number of lags tested. Given $n = 120$ and $k = 3$, critical F -values (5% level of significance) are 3.92 ($p = 1$) and 3.08 ($p = 2$). BG stats in **Indian Equities—Fama-French Model** are lower than the critical F -values; therefore, serial correlation does not seem to be a problem for both lags.

(Module 1.3, LOS 1.i)

Question #74 - 74 of 164

Question ID: 1479948

Should Salve be concerned about residual multicollinearity?

A) Yes, and Salve should exclude variable $R_m - R_f$ from the model.



B) No.



C) Yes, and Salve should exclude either variable SMB or HML from the model.



Explanation

Multicollinearity is detected using the variance inflation factor (VIF). VIF values greater than 5 (i.e., $R^2 > 80\%$) warrant further investigation, while values above 10 (i.e., $R^2 > 90\%$) indicate severe multicollinearity. None of the variables have $VIF > 5$.

(Module 1.3, LOS 1.j)

An analyst is estimating whether company sales is related to three economic variables. The regression exhibits conditional heteroskedasticity, serial correlation, and multicollinearity. The analyst uses White and Newey-West corrected standard errors. Which of the following is *most* accurate?

- A) The regression will still exhibit multicollinearity, but the heteroskedasticity and serial correlation problems will be solved. 
- B) The regression will still exhibit heteroskedasticity and multicollinearity, but the serial correlation problem will be solved. 
- C) The regression will still exhibit serial correlation and multicollinearity, but the heteroskedasticity problem will be solved. 

Explanation

The correction mentioned solves for heteroskedasticity and serial correlation.

(Module 1.3, LOS 1.h)

(Module 1.3, LOS 1.i)

Question #76 of 164

Consider the following analysis of variance (ANOVA) table:

Source	Sum of squares	Degrees of freedom	Mean square
Regression	20	1	20
Error	80	40	2
Total	100	41	

The F-statistic for the test of the fit of the model is *closest* to:

- A) 0.25. 
- B) 0.10. 
- C) 10.00. 

Explanation

The F-statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = \text{MSR}/\text{MSE} = 20 / 2 = 10.$$

(Module 1.2, LOS 1.e)

Question #77 of 164

Question ID: 1472012

Which of the following statements regarding heteroskedasticity is *least* accurate?

- A) Heteroskedasticity may occur in cross-sectional or time-series analyses. 
- B) Heteroskedasticity results in an estimated variance that is too small and, therefore, affects statistical inference. 
- C) The assumption of linear regression is that the residuals are heteroskedastic. 

Explanation

The assumption of regression is that the residuals are homoskedastic (i.e., the residuals are drawn from the same distribution).

(Module 1.3, LOS 1.h)

Question #78 of 164

Question ID: 1479918

During the course of a multiple regression analysis, an analyst has observed several items that she believes may render incorrect conclusions. For example, the coefficient standard errors are too small, although the estimated coefficients are accurate. She believes that these small standard error terms will result in the computed *t*-statistics being too big, resulting in too many Type I errors. The analyst has *most likely* observed which of the following assumption violations in her regression analysis?

- A) Multicollinearity. 
- B) Homoskedasticity. 
- C) Positive serial correlation. 

Explanation

Positive serial correlation is the condition where a positive regression error in one time period increases the likelihood of having a positive regression error in the next time period. The residual terms are correlated with one another, leading to coefficient error terms that are too small.

(Module 1.3, LOS 1.i)

Question #79 of 164

Question ID: 1471882

When interpreting the results of a multiple regression analysis, which of the following terms represents the value of the dependent variable when the independent variables are all equal to zero?

A) Intercept term.



B) p -value.



C) Slope coefficient.



Explanation

The intercept term is the value of the dependent variable when the independent variables are set to zero.

(Module 1.1, LOS 1.b)

Raul Gloucester, CFA, is analyzing the returns of a fund that his company offers. He tests the fund's sensitivity to a small capitalization index and a large capitalization index, as well as to whether the January effect plays a role in the fund's performance. He uses two years of monthly returns data, and runs a regression of the fund's return on the indexes and a January-effect qualitative variable. The "January" variable is 1 for the month of January and zero for all other months. The results of the regression are shown in the tables below.

Regression Statistics	
Multiple R	0.817088
R ²	0.667632
Adjusted R ²	0.617777
Standard Error	1.655891
Observations	24

ANOVA

	df	SS	MS
Regression	3	110.1568	36.71895
Residual	20	54.8395	2.741975
Total	23	164.9963	

	Coefficients	Standard Error	t-Statistic
Intercept	-0.23821	0.388717	-0.61282
January	2.560552	1.232634	2.077301
Small Cap Index	0.231349	0.123007	1.880778
Large Cap Index	0.951515	0.254528	3.738359

Exhibit 1: Partial F-Table (5% Level of Significance)

Degree of Freedom Denominator	Degree of Freedom Numerator		
	1	2	3
18	4.41	3.55	3.16
19	4.38	3.52	3.13
20	4.35	3.49	3.10
21	4.32	3.47	3.07
22	4.30	3.44	3.05
23	4.28	3.42	3.03

Gloucester plans to test for serial correlation and conditional and unconditional heteroskedasticity.

Question #80 - 85 of 164

Question ID: 1479925

The percent of the variation in the fund's return that is explained by the regression is:

A) 66.76%.



B) 81.71%.



C) 61.78%.



Explanation

The R^2 tells us how much of the change in the dependent variable is explained by the changes in the independent variables in the regression: 0.667632.

(Module 1.2, LOS 1.d)

Question #81 - 85 of 164

Question ID: 1479926

Suppose the Breusch-Godfrey statistic is 3.22. At a 5% level of significance, which of the following is the *most* accurate conclusion regarding the presence of serial correlation (at two lags) in the residuals?

- A) No, because the BG statistic is less than the critical test statistic of 3.49, we don't have evidence of serial correlation. 
- B) No, because the BG statistic is less than the critical test statistic of 3.55, we don't have evidence of serial correlation. 
- C) Yes, because the BG statistic exceeds the critical test statistic of 3.16, there is evidence of serial correlation. 

Explanation

Number of lags tested = $p = 2$. The appropriate test statistic for BG test is F -stat with ($p = 2$) and ($n - p - k - 1 = 18$) degrees of freedom. From the table, critical value = 3.55.

(Module 1.3, LOS 1.i)

Question #82 - 85 of 164

Question ID: 1479927

Gloucester subsequently revises the model to exclude the small cap index and finds that the revised model has a RSS of 106.332. Which of the following statements is *most* accurate? At a 5% level of significance, the test statistic:

- A) of 13.39 indicates that we cannot reject the hypothesis that the coefficient of small-cap index is significantly different from 0. 
- B) of 1.30 indicates that we cannot reject the hypothesis that the coefficient of small-cap index is not significantly different from 0. 
- C) of 4.35 indicates that we cannot reject the hypothesis that the coefficient of small-cap index is significantly different from 0. 

Explanation

$$SSE_R = SST - RSS_R = 164.9963 - 106.3320 = 58.6643$$

$$F = [(SSE_R - SSE_U) / q] / [SSE_U / (n - k - 1)] = [(58.6643 - 54.8395) / 1] / (54.8395 / 20) = 3.8248 / 2.742 = 1.30$$

Critical $F_{(1, 20)} = 4.35$ (from Exhibit 1)

Since the test statistic is not greater than the critical value, we cannot reject the null hypothesis that $b_2 = 0$.

(Module 1.2, LOS 1.e)

Question #83 - 85 of 164

Question ID: 1479928

The *best* test for unconditional heteroskedasticity is:

- A) neither the Durbin-Watson test nor the Breusch-Pagan test.
- B) the Breusch-Godfrey test only.
- C) the Breusch-Pagan test only.



Explanation

Breusch-Godfrey and Durbin-Watson tests are for serial correlation. The Breusch-Pagan test is for *conditional* heteroskedasticity; it tests to see if the size of the independent variables influences the size of the residuals. Although tests for unconditional heteroskedasticity exist, they are not part of the CFA curriculum, and unconditional heteroskedasticity is generally considered less serious than conditional heteroskedasticity.

(Module 1.3, LOS 1.h)

(Module 1.3, LOS 1.i)

Question #84 - 85 of 164

Question ID: 1479929

In the month of January, if both the small and large capitalization index have a zero return, we would expect the fund to have a return equal to:

- A) 2.322.
- B) 2.799.
- C) 2.561.



Explanation

The forecast of the return of the fund would be the intercept plus the coefficient on the January effect: $2.322 = -0.238214 + 2.560552$.

(Module 1.2, LOS 1.f)

Question #85 - 85 of 164

Question ID: 1479930

Assuming (for this question only) that the F -test was significant but that the t -tests of the independent variables were insignificant, this would *most likely* suggest:

A) serial correlation.



B) conditional heteroskedasticity.



C) multicollinearity.



Explanation

When the F -test and the t -tests conflict, multicollinearity is indicated.

(Module 1.3, LOS 1.j)

Jessica Jenkins, CFA, is looking at the retail property sector for her manager. She is undertaking a top down review as she feels this is the best way to analyze the industry segment. To predict U.S. property starts (housing), she has used regression analysis.

Jessica included the following variables in her analysis:

- Average nominal interest rates during each year (as a decimal)
- Annual GDP per capita in \$'000

Given these variables, the following output was generated from 30 years of data:

Exhibit 1 – Results from regressing housing starts (in millions) on interest rates and GDP per capita

		Coefficient	Standard Error	T-statistic
Intercept		0.42		3.1
Interest rate		-1.0		-2.0
GDP per capita		0.03		0.7

ANOVA	df	SS	MSS	F
Regression	2	3.896	1.948	21.644
Residual	27	2.431	0.090	
Total	29	6.327		
Observations	30			
Durbin-Watson	1.27			

Exhibit 2 - Critical Values for F-Distribution at 5% Level of Significance

Degrees of Freedom for the Denominator	Degrees of Freedom (df) for the Numerator		
	1	2	3
26	4.23	3.37	2.98
27	4.21	3.35	2.96
28	4.20	3.34	2.95
29	4.18	3.33	2.93
30	4.17	3.32	2.92
31	4.16	3.31	2.91
32	4.15	3.30	2.90

The following variable estimates have been made for 20X7:

GDP per capita = \$46,700

Interest rate = 7%

Question #86 - 88 of 164

Question ID: 1479880

Using the regression model represented in Exhibit 1, what is the predicted number of housing starts for 20X7?

A) 1,751,000.



B) 1,394.



C) 1,394,420.



Explanation

Housing starts = $0.42 - (1 \times 0.07) + (0.03 \times 46.7) = 1.751$ million

(Module 1.2, LOS 1.e)

Question #87 - 88 of 164

Question ID: 1479881

Which of the following statements *best* describes the explanatory power of the estimated regression?

A) The residual standard error of only 0.3 indicates that the regression equation is a good fit for the sample data.



B) The independent variables explain 61.58% of the variation in housing starts.



C) The large *F*-statistic indicates that both independent variables help explain changes in housing starts.



Explanation

The coefficient of determination is the statistic used to identify explanatory power. This can be calculated from the ANOVA table as $3.896 / 6.327 \times 100 = 61.58\%$.

The residual standard error of 0.3 indicates that the standard deviation of the residuals is 0.3 million housing starts. Without knowledge of the data for the dependent variable, it is not possible to assess whether this is a small or a large error.

The *F*-statistic does not enable us to conclude on *both* independent variables. It only allows us to reject the hypothesis that all regression coefficients are zero and accept the hypothesis that at *least one* isn't.

(Module 1.2, LOS 1.d)

(Module 1.2, LOS 1.e)

Question #88 - 88 of 164

Question ID: 1479882

Which of the following is the *least* appropriate statement in relation to *R*-square and adjusted *R*-square?

- A) Adjusted *R*-square decreases when the added independent variable adds little value to the regression model. 
- B) Adjusted *R*-square can be higher than the coefficient of determination for a model with a good fit. 
- C) *R*-square typically increases when new independent variables are added to the regression. 

Explanation

Adjusted *R*-squared cannot exceed *R*-squared (or coefficient of determination) for a multiple regression. .

(Module 1.2, LOS 1.d)

Question #89 of 164

Question ID: 1471927

One of the underlying assumptions of a multiple regression is that the variance of the residuals is constant for various levels of the independent variables. This quality is referred to as:

- A) a linear relationship. 
- B) homoskedasticity. 

C) a normal distribution.



Explanation

Homoskedasticity refers to the basic assumption of a multiple regression model that the variance of the error terms is constant.

(Module 1.1, LOS 1.c)

Question #90 of 164

Question ID: 1479913

One of the main assumptions of a multiple regression model is that the variance of the residuals is constant across all observations in the sample. A violation of the assumption is *most likely* to be described as:

A) heteroskedasticity.



B) positive serial correlation.



C) unstable remnant deviation.



Explanation

Heteroskedasticity is present when the variance of the residuals is not the same across all observations in the sample, and there are sub-samples that are more spread out than the rest of the sample.

(Module 1.3, LOS 1.h)

Question #91 of 164

Question ID: 1472077

Consider the following model of earnings (EPS) regressed against dummy variables for the quarters:

$$EPS_t = \alpha + \beta_1 Q_{1t} + \beta_2 Q_{2t} + \beta_3 Q_{3t}$$

where:

EPS_t is a quarterly observation of earnings per share

Q_{1t} takes on a value of 1 if period t is the second quarter, 0 otherwise

Q_{2t} takes on a value of 1 if period t is the third quarter, 0 otherwise

Q_{3t} takes on a value of 1 if period t is the fourth quarter, 0 otherwise

Which of the following statements regarding this model is *most* accurate? The:

- A) coefficient on each dummy tells us about the difference in earnings per share between the respective quarter and the one left out (first quarter in this case). 
- B) significance of the coefficients cannot be interpreted in the case of dummy variables. 
- C) EPS for the first quarter is represented by the residual. 

Explanation

The coefficients on the dummy variables indicate the difference in EPS for a given quarter, relative to the first quarter.

(Module 1.4, LOS 1.I)

Question #92 of 164

Question ID: 1479901

A multiple regression model has included independent variables that are not linearly related to the dependent variable. The model is *most likely* misspecified due to:

- A) incorrect data pooling. 
- B) incorrect variable scaling. 
- C) incorrect variable form. 

Explanation

Incorrect variable form misspecification occurs if the relationship between dependent and independent variables is nonlinear.

(Module 1.3, LOS 1.g)

Question #93 of 164

Question ID: 1479902

When pooling the samples over multiple economic environments in a multiple regression model, which of the following errors is *most likely* to occur?

- A) Multicollinearity.
- B) Heteroskedasticity.
- C) Model misspecification.



Explanation

When data are improperly pooled over multiple economic environments in a multiple regression analysis, the model would be misspecified.

(Module 1.3, LOS 1.g)

A real estate agent wants to develop a model to predict the selling price of a home. The agent believes that the most important variables in determining the price of a house are its size (in square feet) and the number of bedrooms. Accordingly, he takes a random sample of 32 homes that has recently been sold. The results of the regression are:

	Coefficient	Standard Error	t-statistics
Intercept	66,500	59,292	1.12
House Size	74.30	21.11	3.52
Number of Bedrooms	10306	3230	3.19

$$R^2 = 0.56; F = 40.73$$

Selected F- table values for significance level of 0.05:

	1	2
28	4.20	3.34
29	4.18	3.33
30	4.17	3.32
32	4.15	3.29

(Degrees of freedom for the numerator in columns; Degrees of freedom for the denominator in rows)

Additional information regarding this multiple regression:

1. Variance of error is not constant across the 32 observations.
2. The two variables (size of the house and the number of bedrooms) are highly correlated.
3. The error variance is not correlated with the size of the house nor with the number of bedrooms.

Question #94 - 96 of 164

Question ID: 1479936

The predicted price of a house that has 2,000 square feet of space and 4 bedrooms is *closest* to:

A) \$114,000.



B) \$256,000.



C) \$185,000.



Explanation

$$66,500 + 74.30(2,000) + 10,306(4) = \$256,324$$

(Module 1.2, LOS 1.f)

Question #95 - 96 of 164

Question ID: 1479937

The conclusion from the hypothesis test of $H_0: b_1 = b_2 = 0$, is that the null hypothesis should:

A) not be rejected as the calculated F of 40.73 is greater than the critical value of 3.29.



B) be rejected as the calculated F of 40.73 is greater than the critical value of 3.33.



C) be rejected as the calculated F of 40.73 is greater than the critical value of 3.29.



Explanation

We can reject the null hypothesis that coefficients of both independent variables equal 0. The F value for comparison is $F_{2,29} = 3.33$. The degrees of freedom in the numerator is 2; equal to the number of independent variables. Degrees of freedom for the denominator is $32 - (2+1) = 29$. The critical value of the F-test needed to reject the null hypothesis is thus 3.33. The actual value of the F-test statistic is 40.73, so the null hypothesis should be rejected, as the calculated F of 40.73 is greater than the critical value of 3.33.

(Module 1.2, LOS 1.e)

Question #96 - 96 of 164

Question ID: 1479938

Which of the following is *most likely* to present a problem in using this regression for forecasting?

A) Heteroskedasticity.



B) Multicollinearity.



C) Autocorrelation.



Explanation

Multicollinearity is present in a regression model when some linear combination of the independent variables are highly correlated. We are told that the two independent variables in this question are highly correlated. We also recognize that unconditional heteroskedasticity is present – but this would not pose any major problems in using this model for forecasting. No information is given about autocorrelation in residuals, but this is generally a concern with time series data (in this case, the model uses cross-sectional data).

(Module 1.3, LOS 1.j)

Som Muttney has been asked to forecast the level of operating profit for a proposed new branch of a tire store. His forecast is one component in forecasting operating profit for the entire company for the next fiscal year. Muttney decide to conduct multiple regression analysis using "branch store operating profit" as the dependent variable and three independent variables. The three independent variables are "population within 5 miles of the branch," "operating hours per week," and "square footage of the facility." Muttney used data on the company's existing 23 branches to develop the model ($n=23$).

Regression of Operating Profit on Population, Operating Hours, and Square Footage

--	--

Dependent Variable	Operating Profit (Y)	
Independent Variables	Coefficient Estimate	t-value
Intercept	103,886	2.740
Population within 5 miles (X_1)	4.372	2.133
Operating hours per week (X_2)	214.856	0.258
Square footage of facility (X_3)	6.767	2.643
Regression sum of squares	6,349	
Sum of squares total	10,898	

	Two-tailed Significance				
<i>Degrees of Freedom</i>	.20	.10	.05	.02	.01
3	1.638	2.353	3.182	4.541	5.841
19	1.328	1.729	2.093	2.539	2.861
23	1.319	1.714	2.069	2.50	2.807

In his research report, Muttney claims that when the square footage of the store is increased by 1%, operating profit will increase by more than 5%

Question #97 - 102 of 164

Question ID: 1471921

The 95% confidence interval for slope coefficient for independent variable "population" is *closest* to:

A) 0.081 – 8.66



B) -0.81 – 9.56



C) -0.086 – 8.83



Explanation

The degrees of freedom are $[n - k - 1]$. Here, n is the number of observations in the regression (23) and k is the number of independent variables (3). $df = [23 - 3 - 1] = 19$. t_c (for $\alpha = 5/2 = 2.5\%$) = 2.093.

$$S_e \text{ (beta for population)} = \text{beta}/t\text{-value} = 4.372 / 2.133 = 2.05$$

$$95\% \text{ confidence interval} = \text{Coefficient} \pm t_c \times S_e = 4.372 \pm 2.093 \times 2.05 = 0.08135 - 8.66265$$

(Module 1.1, LOS 1.b)

Question #98 - 102 of 164

Question ID: 1471922

The probability of finding a value of t for variable X_1 that is as-large or larger than $|2.133|$ when the null hypothesis is true is:

- A) between 5% and 10%.
- B) between 1% and 2%.
- C) between 2% and 5%.



Explanation

The degrees of freedom is:

$$\begin{aligned} &= (n - k - 1) \\ &= (23 - 3 - 1) \\ &= 19 \end{aligned}$$

In the table above, for 19 degrees of freedom, the value 2.133 would lie between a 2% chance (alpha of 0.02) or 2.539 and a 5% chance (alpha of 0.05) or 2.093.

(Module 1.1, LOS 1.b)

Question #99 - 102 of 164

Question ID: 1471923

The correlation between the actual values of operating profit and the predicted value of operating profit is *closest* to:

- A) 0.53
- B) 0.76
- C) 0.36



Explanation

$R^2 = \text{RSS}/\text{SST} = 6,349/10,898 = 0.58$. Correlation between predicted and actual values of dependent variable $= (0.58)^{0.5} = 0.76$

(Module 1.1, LOS 1.b)

Question #100 - 102 of 164

Question ID: 1471924

Regarding Muttney's claim about a 5% increase in operating profit for a 1% increase in square footage, the most appropriate null hypothesis and conclusion (at a 5% level of significance) are:

	<u>Null Hypothesis</u>	<u>Conclusion</u>	
A)	$H_0: b_3 \geq 5$	Fail to reject H_0	
B)	$H_0: b_3 \leq 5$	Reject H_0	
C)	$H_0: b_3 \leq 5$	Fail to reject H_0	

Explanation

S_e (beta for sq footage) $= \text{beta}/t\text{-value} = 6.767/2.643 = 2.56$

$t_c(\alpha = 5\%, \text{one-tailed, dof} = 19) = 1.729$

$t = \text{beta} - \text{beta}_0/S_e = 6.767 - 5/2.56 = 0.69$. We fail to reject the null hypothesis

(Module 1.1, LOS 1.b)

Question #101 - 102 of 164

Question ID: 1471925

The standard deviation of regression residuals is *closest* to:

- | | |
|-----------|---|
| A) 239.42 |  |
| B) 15.47 |  |
| C) 0.42 |  |

Explanation

$$SSE = SST - RSS = 10,898 - 6,349 = 4,549$$

$$MSE = SSE/(n-k-1) = 4,549/19 = 239.42$$

$$SEE = (MSE)^{0.5} = 15.47$$

(Module 1.1, LOS 1.b)

Question #102 - 102 of 164

Question ID: 1471926

The operating profit model as specified is *most likely* a:

- A) Time series regression
- B) Autoregressive model
- C) Cross-sectional regression



Explanation

Cross-sectional data involve many observations for the same time period. Time-series data uses many observations from different time periods for the same entity.

(Module 1.1, LOS 1.b)

Question #103 of 164

Question ID: 1471870

Consider the following regression equation:

$$\text{Sales}_i = 10.0 + 1.25 \text{ R\&D}_i + 1.0 \text{ ADV}_i - 2.0 \text{ COMP}_i + 8.0 \text{ CAP}_i$$

where Sales is dollar sales in millions, R&D is research and development expenditures in millions, ADV is dollar amount spent on advertising in millions, COMP is the number of competitors in the industry, and CAP is the capital expenditures for the period in millions of dollars.

Which of the following is NOT a correct interpretation of this regression information?

- If R&D and advertising expenditures are \$1 million each, there are 5
- A) competitors, and capital expenditures are \$2 million, expected Sales are \$8.25 million.



- B)** One more competitor will mean \$2 million less in Sales (holding everything else constant). 
- C)** If a company spends \$1 million more on capital expenditures (holding everything else constant), Sales are expected to increase by \$8.0 million. 

Explanation

Predicted sales = $\$10 + 1.25 + 1 - 10 + 16 = \18.25 million.

(Module 1.1, LOS 1.b)

Question #104 of 164

Question ID: 1471928

Which of the following statements *least* accurately describes one of the fundamental multiple regression assumptions?

- A)** The error term is normally distributed. 
- B)** The variance of the error terms is not constant (i.e., the errors are heteroskedastic). 
- C)** The independent variables are not random. 

Explanation

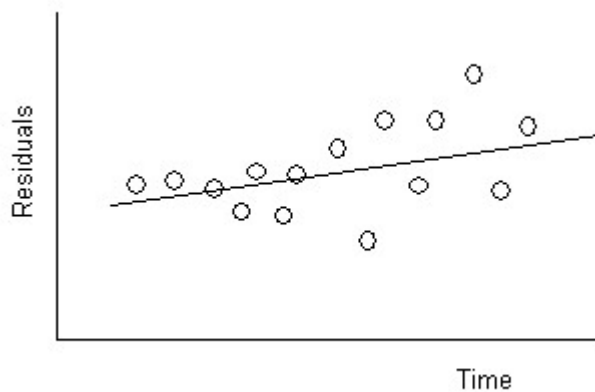
The variance of the error term IS assumed to be constant, resulting in errors that are homoskedastic.

(Module 1.1, LOS 1.c)

Question #105 of 164

Question ID: 1472011

Consider the following graph of residuals and the regression line from a time-series regression:



These residuals exhibit the regression problem of:

- A) homoskedasticity.
- B) autocorrelation.
- C) heteroskedasticity.



Explanation

The residuals appear to be from two different distributions over time. In the earlier periods, the model fits rather well compared to the later periods.

(Module 1.3, LOS 1.h)

Question #106 of 164

Question ID: 1471869

Consider the following regression equation:

$$\text{Sales}_i = 20.5 + 1.5 \text{ R\&D}_i + 2.5 \text{ ADV}_i - 3.0 \text{ COMP}_i$$

where Sales is dollar sales in millions, R&D is research and development expenditures in millions, ADV is dollar amount spent on advertising in millions, and COMP is the number of competitors in the industry.

Which of the following is NOT a correct interpretation of this regression information?

- A) If R&D and advertising expenditures are \$1 million each and there are 5 competitors, expected sales are \$9.5 million.
- B) One more competitor will mean \$3 million less in sales (holding everything else constant).



- C) If a company spends \$1 more on R&D (holding everything else constant), sales are expected to increase by \$1.5 million.



Explanation

If a company spends \$1 *million* more on R&D (holding everything else constant), sales are expected to increase by \$1.5 million. Always be aware of the units of measure for the different variables.

(Module 1.1, LOS 1.b)

Question #107 of 164

Question ID: 1479908

An analyst is trying to determine whether fund return performance is persistent. The analyst divides funds into three groups based on whether their return performance was in the top third (group 1), middle third (group 2), or bottom third (group 3) during the previous year. The manager then creates the following equation: $R = a + b_1D_1 + b_2D_2 + b_3D_3 + \varepsilon$, where R is return premium on the fund (the return minus the return on the S&P 500 benchmark) and D_i is equal to 1 if the fund is in group i . Assuming no other information, this equation will suffer from:

- A) multicollinearity.
- B) heteroskedasticity.
- C) serial correlation.



Explanation

When we use dummy variables, we have to use one less than the states of the world. In this case, there are three states (groups) possible. We should have used only two dummy variables. Multicollinearity is a problem in this case. Specifically, a linear combination of independent variables is perfectly correlated. $X_1 + X_2 + X_3 = 1$.

There are too many dummy variables specified, so the equation will suffer from multicollinearity.

(Module 1.3, LOS 1.h)

Question #108 of 164

Question ID: 1479867

Jason Fye, CFA, wants to check for seasonality in monthly stock returns (i.e., the January effect) after controlling for market cap and systematic risk. The type of model that Fye would *most* appropriately select is:

A) Neither multiple regression nor logistic regression.



B) logistic regression model.



C) Multiple regression model.



Explanation

Fye wants to test a theory of January effect on stock returns (dependent variable) using a dummy (January = 1, other months = 0), market cap, and beta (independent variables). A multiple regression model would be most appropriate. Because the dependent variable (stock returns) is not a qualitative variable, a logistic regression would not apply.

(Module 1.1, LOS 1.a)

Question #109 of 164

Question ID: 1471871

Henry Hilton, CFA, is undertaking an analysis of the bicycle industry. He hypothesizes that bicycle sales (SALES) are a function of three factors: the population under 20 (POP), the level of disposable income (INCOME), and the number of dollars spent on advertising (ADV). All data are measured in millions of units. Hilton gathers data for the last 20 years. Which of the follow regression equations *correctly* represents Hilton's hypothesis?

A) $\text{INCOME} = \alpha + \beta_1 \text{POP} + \beta_2 \text{SALES} + \beta_3 \text{ADV} + \epsilon$.



B) $\text{SALES} = \alpha + \beta_1 \text{POP} + \beta_2 \text{INCOME} + \beta_3 \text{ADV} + \epsilon$.



C) $\text{SALES} = \alpha \times \beta_1 \text{POP} \times \beta_2 \text{INCOME} \times \beta_3 \text{ADV} \times \epsilon$.



Explanation

SALES is the dependent variable. POP, INCOME, and ADV should be the independent variables (on the right hand side) of the equation (in any order). Regression equations are additive.

(Module 1.1, LOS 1.b)

Question #110 of 164

Question ID: 1472026

Which of the following statements regarding serial correlation that might be encountered in regression analysis is *least* accurate?

A) Positive serial correlation and heteroskedasticity can both lead to Type I errors.



B) Serial correlation occurs least often with time series data.



C) Serial correlation does not affect consistency of regression coefficients.



Explanation

Serial correlation, which is sometimes referred to as autocorrelation, occurs when the residual terms are correlated with one another, and is most frequently encountered with time series data. Positive serial correlation can lead to standard errors that are too small, which will cause computed t-statistics to be larger than they should be, which will lead to too many Type I errors (i.e. the rejection of the null hypothesis when it is actually true). Serial correlation however does not affect the consistency of the regression coefficients.

(Module 1.3, LOS 1.h)

Question #111 of 164

Question ID: 1471945

Which of the following statements regarding the R^2 is *least* accurate?

- A) The F-statistic for the test of the fit of the model is the ratio of the mean squared regression to the mean squared error. 
- B) The R^2 of a regression will be greater than or equal to the adjusted- R^2 for the same regression. 
- C) The R^2 is the ratio of the unexplained variation to the explained variation of the dependent variable. 

Explanation

The R^2 is the ratio of the explained variation to the total variation.

(Module 1.2, LOS 1.d)

Werner Baltz, CFA, has regressed 30 years of data to forecast future sales for National Motor Company based on the percent change in gross domestic product (GDP) and the change in retail price of a U.S. gallon of fuel. The results are presented below.

Predictor	Coefficient	Standard Error of the Coefficient
Intercept	78	13.710
Δ GDP	30.22	12.120
Δ \$ Fuel	-412.39	183.981

Analysis of Variance Table (ANOVA)

Source	Degrees of Freedom	Sum of Squares
Regression		291.30
Error	27	132.12
Total	29	423.42

Question #112 - 114 of 164

Question ID: 1479910

If GDP rises 2.2% and the price of fuels falls \$0.15, Baltz's model will predict Company sales to be (in \$ millions) *closest* to:

- A) \$82.00. ✗
- B) \$128.00. ✗
- C) \$206.00. ✓

Explanation

Sales will be closest to $\$78 + (\$30.22 \times 2.2) + [(-412.39) \times (-\$0.15)] = \$206.34$ million.

(Module 1.2, LOS 1.f)

Question #113 - 114 of 164

Question ID: 1479911

Baltz proceeds to test the hypothesis that none of the independent variables has significant explanatory power. He concludes that, at a 5% level of significance:

- A) all of the independent variables have explanatory power, because the calculated F -statistic exceeds its critical value. ✗
- B) none of the independent variables has explanatory power, because the calculated F -statistic does not exceed its critical value. ✗
- C) at least one of the independent variables has explanatory power, because the calculated F -statistic exceeds its critical value. ✓

Explanation

$MSE = SSE / [n - (k + 1)] = 132.12 \div 27 = 4.89$. From the ANOVA table, the calculated F -statistic is (mean square regression / mean square error) = $145.65 / 4.89 = 29.7853$. From the F -distribution table (2 df numerator, 27 df denominator) the F -critical value may be interpolated to be 3.36. Because 29.7853 is greater than 3.36, Baltz rejects the null hypothesis and concludes that at least one of the independent variables has explanatory power.

(Module 1.2, LOS 1.e)

Question #114 - 114 of 164

Question ID: 1479912

Presence of conditional heteroskedasticity is *least likely* to affect the:

A) computed F -statistic.



B) coefficient estimates.



C) computed t -statistic.



Explanation

Conditional heteroskedasticity results in consistent coefficient estimates, but it biases standard errors, affecting the computed t -statistic and F -statistic.

(Module 1.3, LOS 1.h)

Question #115 of 164

Question ID: 1479878

Consider the following analysis of variance table:

Source	Sum of Squares	Df	Mean Square
Regression	20	1	20
Error	80	20	4
Total	100	21	

The F -statistic for a test of joint significance of all the slope coefficients is *closest* to:

A) 0.2.



B) 5.



C) 0.05.



Explanation

The F -statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = MSR / MSE = 20 / 4 = 5.$$

(Module 1.2, LOS 1.e)

Manuel Mercado, CFA has performed the following two regressions on sales data for a given industry. He wants to forecast sales for each quarter of the upcoming year.

Model ONE	
Regression Statistics	
Multiple R	0.941828
R ₂	0.887039
Adjusted R ₂	0.863258
Standard Error	2.543272
Observations	24

Durbin-Watson test statistic = 0.7856

ANOVA					
	df	SS	MS	F	Significance F
Regression	4	965.0619	241.2655	37.30006	9.49E-09
Residual	19	122.8964	6.4682		
Total	23	1087.9583			

	Coefficients	Standard Error	t-Statistic
Intercept	31.40833	1.4866	21.12763
Q1	-3.77798	1.485952	-2.54246
Q2	-2.46310	1.476204	-1.66853
Q3	-0.14821	1.470324	-0.10080
TREND	0.851786	0.075335	11.20848

Model TWO	
Regression Statistics	
Multiple R	0.941796
R ²	0.886979
Adjusted R ²	0.870026
Standard Error	2.479538
Observations	24

Durbin-Watson test statistic = 0.7860

	df	SS	MS	F	Significance F
Regression	3	964.9962	321.6654	52.3194	1.19E-09
Residual	20	122.9622	6.14811		
Total	23	1087.9584			

	Coefficients	Standard Error	t-Statistic
Intercept	31.32888	1.228865	25.49416
Q1	-3.70288	1.253493	-2.95405
Q2	-2.38839	1.244727	-1.91881
TREND	0.85218	0.073991	11.51732

The dependent variable is the level of sales for each quarter, in \$ millions, which began with the first quarter of the first year. Q1, Q2, and Q3 are seasonal dummy variables representing each quarter of the year. For the first four observations the dummy variables are as follows: Q1:(1,0,0,0), Q2:(0,1,0,0), Q3:(0,0,1,0). The TREND is a series that begins with one and increases by one each period to end with 24. For all tests, Mercado will use a 5% level of significance. Tests of coefficients will be two-tailed, and all others are one-tailed.

Question #116 - 121 of 164

Question ID: 1471938

Which model would be a better choice for making a forecast?

A) Model TWO because it has a higher adjusted R².



B) Model ONE because it has a higher R².



C) Model TWO because serial correlation is not a problem.



Explanation

Model TWO has a higher adjusted R^2 and thus would produce the more reliable estimates. As is always the case when a variable is removed, R^2 for Model TWO is lower. The increase in adjusted R^2 indicates that the removed variable, Q3, has very little explanatory power, and removing it should improve the accuracy of the estimates. With respect to the references to autocorrelation, we can compare the Durbin-Watson statistics to the critical values on a Durbin-Watson table. Since the critical DW statistics for Model ONE and TWO respectively are 1.01 (>0.7856) and 1.10 (>0.7860), serial correlation is a problem for both equations.

(Module 1.2, LOS 1.d)

Question #117 - 121 of 164

Question ID: 1479951

Using Model ONE, what is the sales forecast for the second quarter of the next year?

A) \$56.02 million.



B) \$46.31 million.



C) \$51.09 million.



Explanation

The estimate for the second quarter of the following year would be (in millions):

$$31.4083 + (-2.4631) + (24 + 2) \times 0.851786 = 51.091666.$$

(Module 1.2, LOS 1.f)

Question #118 - 121 of 164

Question ID: 1479952

Which model misspecification is *most likely* to cause multicollinearity?

A) Inappropriate variable scaling.



B) Omission of important variable(s).



C) Inappropriate variable form.



Explanation

Inappropriate variable scaling may lead to multicollinearity or heteroskedasticity in residuals. Omission of important variable may lead to biased and inconsistent regression parameters and also heteroskedasticity/serial correlation in residuals. Inappropriate variable form can lead to heteroskedasticity in residuals.

(Module 1.3, LOS 1.g)

Question #119 - 121 of 164

Question ID: 1479953

If it is determined that conditional heteroskedasticity is present in model one, which of the following inferences are *most* accurate?

- A) Regression coefficients will be unbiased but standard errors will be biased. 
- B) Both the regression coefficients and the standard errors will be biased. 
- C) Regression coefficients will be biased but standard errors will be unbiased. 

Explanation

Presence of conditional heteroskedasticity will not affect the consistency of regression coefficients but will bias the standard errors leading to incorrect application of t-tests for statistical significance of regression parameters.

(Module 1.3, LOS 1.h)

Question #120 - 121 of 164

Question ID: 1479954

Mercado probably did not include a fourth dummy variable Q4, which would have had 0, 0, 0, 1 as its first four observations because:

- A) the intercept is essentially the dummy for the fourth quarter. 
- B) it would not have been significant. 
- C) it would have lowered the explanatory power of the equation. 

Explanation

The fourth quarter serves as the base quarter, and for the fourth quarter, $Q1 = Q2 = Q3 = 0$. Had the model included a Q4 as specified, we could not have had an intercept. In that case, for Model ONE for example, the estimate of Q4 would have been 31.40833. The dummies for the other quarters would be the 31.40833 plus the estimated dummies from the Model ONE. In a model that included Q1, Q2, Q3, and Q4 but no intercept, for example:

$$Q1 = 31.40833 + (-3.77798) = 27.63035$$

Such a model would produce the same estimated values for the dependent variable.

(Module 1.4, LOS 1.I)

Question #121 - 121 of 164

Question ID: 1479955

If Mercado determines that Model TWO is the appropriate specification, then he is essentially saying that for each year, value of sales from quarter three to four is expected to:

- A) grow by more than \$1,000,000.
- B) grow, but by less than \$1,000,000.
- C) remain approximately the same.



Explanation

The specification of Model TWO essentially assumes there is no difference attributed to the change of the season from the third to fourth quarter. However, the time trend is significant. The trend effect for moving from one season to the next is the coefficient on TREND times \$1,000,000 which is \$852,182 for Equation TWO.

(Module 1.1, LOS 1.b)

Vijay Shapule, CFA, is investigating the application of the Fama-French three-factor model (Model 1) for the Indian stock market for the period 2001–2011 (120 months). Using the dependent variable as annualized return (%), the results of the analysis are shown in **Indian Equities—Fama-French Model**.

Indian Equities—Fama-French Model

Factor	Coefficient	P-Value
Intercept	1.22	<0.001
SMB	0.23	<0.001
HML	0.34	0.003
Rm-Rf	0.88	<0.001

R-squared	0.36
SSE	38.00
AIC	-129.99
BIC	-118.84

Shapule then modifies the model to include a liquidity factor. Results for this four-factor model (Model 2) are shown in **Revised Fama-French Model With Liquidity Factor**

Revised Fama-French Model With Liquidity Factor

Factor	Coefficient	P-Value
Intercept	1.56	<0.001
SMB	0.22	<0.001
HML	0.35	0.012
Rm-Rf	0.87	<0.001
LIQ	-0.12	0.02
R-squared	0.39	
SSE	34.00	
AIC	-141.34	
BIC	-127.40	

Question #122 - 125 of 164

Question ID: 1479897

The adjusted R^2 of Model 2 is *closest* to:

- A) 0.36.
- B) 0.39.
- C) 0.37.



Explanation

Given $n = 120$ months, $k = 4$ (for Model 2), and $R^2 = 0.39$:

$$R_a^2 = 1 - \left[\left(\frac{120-1}{120-4-1} \right) \times (1 - 0.39) \right] = 0.37$$

(Module 1.2, LOS 1.d)

Question #123 - 125 of 164

Question ID: 1479898

The model better suited for prediction is:

- A) Model 2 because it has a lower Akaike information criterion.
- B) Model 1 because it has a lower Bayesian information criterion.
- C) Model 2 because it has a higher Akaike information criterion.

**Explanation**

The Akaike information criterion (AIC) is used if the goal is to have a better forecast, while the Bayesian information criterion (BIC) is used if the goal is a better goodness of fit. Lower values of both criteria indicate a better model. Both criteria are lower for Model 2.

(Module 1.2, LOS 1.d)

Question #124 - 125 of 164

Question ID: 1479899

The F -statistic for testing H_0 : coefficient of LIQ = 0 versus H_a : coefficient of LIQ $\neq$ 0 is *closest* to:

- A) 2.11.
- B) 13.33.
- C) 5.45.

**Explanation**

$$F = \frac{(SSE_R - SSE_U)/q}{(SSE_U)/(n-k-1)}$$

where $n = 120$, $k = 4$, and $q = 1$:

$$\frac{(38-34)/1}{(34)/(120-4-1)} = 13.33$$

(Module 1.2, LOS 1.d)

Question #125 - 125 of 164

Question ID: 1479900

What is the predicted return for a stock using Model 1 when SMB = 3.30, HML = 1.25 and $R_m - R_f = 5$?

A) 9.58%.



B) 6.80%.



C) 7.88%.



Explanation

Model 1:

$$\begin{aligned}\text{Return} &= 1.22 + 0.23 \times \text{SMB} + 0.34 \times \text{HML} + 0.88 R_m - R_f \\ &= 1.22 + 0.23 \times 3.30 + 0.34 \times 1.25 + 0.88 \times 5 \\ &= 6.80\%.\end{aligned}$$

(Module 1.2, LOS 1.d)

Question #126 of 164

Question ID: 1479923

Assume that in a particular multiple regression model, it is determined that the error terms are uncorrelated with each other. Which of the following statements is *most* accurate?

A) Serial correlation may be present in this multiple regression model, and can be confirmed only through a Durbin-Watson test.



B) Unconditional heteroskedasticity present in this model should not pose a problem, but can be corrected by using robust standard errors.



C) This model is in accordance with the basic assumptions of multiple regression analysis because the errors are not serially correlated.



Explanation

One of the basic assumptions of multiple regression analysis is that the error terms are not correlated with each other. In other words, the error terms are not serially correlated. Multicollinearity and heteroskedasticity are problems in multiple regression that are not related to the correlation of the error terms.

(Module 1.3, LOS 1.i)

Question #127 of 164

Question ID: 1479921

Alex Wade, CFA, is analyzing the result of a regression analysis comparing the performance of gold stocks versus a broad equity market index. Wade believes that first lag serial correlation may be present and, in order to prove his theory, should use which of the following methods to detect its presence?

A) The Durbin-Watson statistic.



B) The Breusch-Pagan test.



C) The Hansen method.



Explanation

The Durbin-Watson statistic is the most commonly used method for the detection of serial correlation at the first lag, although residual plots can also be utilized. For testing of serial correlation beyond the first lag, we can instead use the Breusch-Godfrey test (but is not one of the answer choices).

(Module 1.3, LOS 1.i)

Question #128 of 164

Question ID: 1479873

May Jones estimated a regression that produced the following analysis of variance (ANOVA) table:

Source	Sum of squares	Degrees of freedom	Mean square
Regression	20	1	20
Error	80	40	2
Total	100	41	

The values of R^2 and the F -statistic for joint test of significance of all the slope coefficients are:

A) $R^2 = 0.25$ and $F = 0.909$.



B) $R^2 = 0.20$ and $F = 10$.



C) $R^2 = 0.25$ and $F = 10$.



Explanation

$$R^2 = \text{RSS} / \text{SST} = 20 / 100 = 0.20$$

The F -statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = 20 / 2 = 10$$

(Module 1.2, LOS 1.d)

(Module 1.2, LOS 1.e)

Question #129 of 164

Question ID: 1471872

Henry Hilton, CFA, is undertaking an analysis of the bicycle industry. He hypothesizes that bicycle sales (SALES) are a function of three factors: the population under 20 (POP), the level of disposable income (INCOME), and the number of dollars spent on advertising (ADV). All data are measured in millions of units. Hilton gathers data for the last 20 years and estimates the following equation (standard errors in parentheses):

SALES	= α	+ 0.004 POP	+ 1.031 INCOME	+ 2.002 ADV
		(0.005)	(0.337)	(2.312)

The critical t -statistic for a 95% confidence level is 2.120. Which of the independent variables is statistically different from zero at the 95% confidence level?

A) INCOME and ADV.



B) ADV only.



C) INCOME only.



Explanation

The calculated test statistic is coefficient/standard error. Hence, the t -stats are 0.8 for POP, 3.059 for INCOME, and 0.866 for ADV. Since the t -stat for INCOME is the only one greater than the critical t -value of 2.120, only INCOME is significantly different from zero.

(Module 1.1, LOS 1.b)

Question #130 of 164

Question ID: 1472073

A fund has changed managers twice during the past 10 years. An analyst wishes to measure whether either of the changes in managers has had an impact on performance. R is the return on the fund, and M is the return on a market index. Which of the following regression equations can appropriately measure the desired impacts?

A) The desired impact cannot be measured. 

$R = a + bM + c_1D_1 + c_2D_2 + c_3D_3 + \varepsilon$, where $D_1 = 1$ if the return is from the first

B) manager, and $D_2 = 1$ if the return is from the second manager, and $D_3 = 1$ is the 

return is from the third manager.

$R = a + bM + c_1D_1 + c_2D_2 + \varepsilon$, where $D_1 = 1$ if the return is from the first

C) manager, and $D_2 = 1$ if the return is from the third manager. 

Explanation

The effect needs to be measured by two distinct dummy variables. The use of three variables will cause collinearity, and the use of one dummy variable will not appropriately specify the manager impact.

(Module 1.4, LOS 1.I)

Dave Turner is a security analyst who is using regression analysis to determine how well two factors explain returns for common stocks. The independent variables are the natural logarithm of the number of analysts following the companies, $\ln(\text{no. of analysts})$, and the natural logarithm of the market value of the companies, $\ln(\text{market value})$. The regression output generated from a statistical program is given in the following tables. Each p-value corresponds to a two-tail test.

Turner plans to use the result in the analysis of two investments. WLK Corp. has twelve analysts following it and a market capitalization of \$2.33 billion. NGR Corp. has two analysts following it and a market capitalization of \$47 million.

Table 1: Regression Output

Variable	Coefficient	Standard Error of the Coefficient	t-statistic	p-value
Intercept	0.043	0.01159	3.71	< 0.001
$\ln(\text{No. of Analysts})$	-0.027	0.00466	-5.80	< 0.001
$\ln(\text{Market Value})$	0.006	0.00271	2.21	0.028

Table 2: ANOVA

	Degrees of Freedom	Sum of Squares	Mean Square
Regression	2	0.103	0.051
Residual	194	0.559	0.003
Total	196	0.662	

Question #131 - 136 of 164

Question ID: 1471885

In a one-sided test and a 1% level of significance, which of the following coefficients is significantly different from zero?

- A) The intercept and the coefficient on $\ln(\text{no. of analysts})$ only.
- B) The intercept and the coefficient on $\ln(\text{market value})$ only.
- C) The coefficient on $\ln(\text{no. of Analysts})$ only.



Explanation

The p-values correspond to a two-tail test. For a one-tailed test, divide the provided p-value by two to find the minimum level of significance for which a null hypothesis of a coefficient equaling zero can be rejected. Dividing the provided p-value for the intercept and $\ln(\text{no. of analysts})$ will give a value less than 0.0005, which is less than 1% and would lead to a rejection of the hypothesis. Dividing the provided p-value for $\ln(\text{market value})$ will give a value of 0.014 which is greater than 1%; thus, that coefficient is not significantly different from zero at the 1% level of significance.

(Module 1.1, LOS 1.b)

Question #132 - 136 of 164

Question ID: 1471886

The 95% confidence interval (use a t-stat of 1.96 for this question only) of the estimated coefficient for the independent variable $\ln(\text{Market Value})$ is *closest* to:

- A) 0.014 to -0.009.
- B) -0.018 to -0.036.
- C) 0.011 to 0.001.



Explanation

The confidence interval is $0.006 \pm (1.96)(0.00271) = 0.011 \text{ to } 0.001$

(Module 1.1, LOS 1.b)

Question #133 - 136 of 164

Question ID: 1471887

If the number of analysts on NGR Corp. were to double to 4, the change in the forecast of NGR would be *closest* to?

A) -0.055.



B) -0.019.



C) -0.035.



Explanation

Initially, the estimate is $0.1303 = 0.043 + \ln(2)(-0.027) + \ln(47000000)(0.006)$

Then, the estimate is $0.1116 = 0.043 + \ln(4)(-0.027) + \ln(47000000)(0.006)$

$0.1116 - 0.1303 = -0.0187$, or -0.019

(Module 1.1, LOS 1.b)

Question #134 - 136 of 164

Question ID: 1471888

Based on a R^2 calculated from the information in Table 2, the analyst should conclude that the number of analysts and $\ln(\text{market value})$ of the firm explain:

A) 18.4% of the variation in returns.



B) 84.4% of the variation in returns.



C) 15.6% of the variation in returns.



Explanation

R^2 is the percentage of the variation in the dependent variable (in this case, variation of returns) explained by the set of independent variables. R^2 is calculated as follows: $R^2 = (SSR / SST) = (0.103 / 0.662) = 15.6\%$.

(Module 1.1, LOS 1.b)

Question #135 - 136 of 164

Question ID: 1471889

What is the F-statistic from the regression? And, what can be concluded from its value at a 1% level of significance?

- F = 1.97, fail to reject a hypothesis that both of the slope coefficients are equal to zero. ✗
- A) ✗
- B) F = 5.80, reject a hypothesis that both of the slope coefficients are equal to zero. ✗
- C) F = 17.00, reject a hypothesis that both of the slope coefficients are equal to zero. ✓

Explanation

The F-statistic is calculated as follows: $F = MSR / MSE = 0.051 / 0.003 = 17.00$; and $17.00 > 4.61$, which is the critical F-value for the given degrees of freedom and a 1% level of significance. However, when F-values are in excess of 10 for a large sample like this, a table is not needed to know that the value is significant.

(Module 1.1, LOS 1.b)

Question #136 - 136 of 164

Question ID: 1471890

Upon further analysis, Turner concludes that multicollinearity is a problem. What might have prompted this further analysis and what is intuition behind the conclusion?

- A) At least one of the t -statistics was not significant, the F-statistic was significant, and an intercept not significantly different from zero would be expected. ✗
- B) At least one of the t -statistics was not significant, the F-statistic was not significant, and a positive relationship between the number of analysts and the size of the firm would be expected. ✗
- C) At least one of the t -statistics was not significant, the F-statistic was significant, and a positive relationship between the number of analysts and the size of the firm would be expected. ✓

Explanation

Multicollinearity occurs when there is a high correlation among independent variables and may exist if there is a significant F-statistic for the fit of the regression model, but at least one insignificant independent variable when we expect all of them to be significant. In this case the coefficient on $\ln(\text{market value})$ was not significant at the 1% level, but the F-statistic was significant. It would make sense that the size of the firm, i.e., the market value, and the number of analysts would be positively correlated.

(Module 1.1, LOS 1.b)

Question #137 of 164

Question ID: 1471981

Consider the following estimated regression equation, with standard errors of the coefficients as indicated:

$$\text{Sales}_i = 10.0 + 1.25 \text{ R\&D}_i + 1.0 \text{ ADV}_i - 2.0 \text{ COMP}_i + 8.0 \text{ CAP}_i$$

where the standard error for R&D is 0.45, the standard error for ADV is 2.2, the standard error for COMP 0.63, and the standard error for CAP is 2.5.

Sales are in millions of dollars. An analyst is given the following predictions on the independent variables: R&D = 5, ADV = 4, COMP = 10, and CAP = 40.

The predicted level of sales is *closest* to:

- A) \$320.25 million.
- B) \$310.25 million.
- C) \$300.25 million.



Explanation

Predicted sales

$$= \$10 + 1.25 (5) + 1.0 (4) - 2.0 (10) + 8 (40)$$

$$= 10 + 6.25 + 4 - 20 + 320 = \$320.25$$

(Module 1.2, LOS 1.f)

Question #138 of 164

Question ID: 1479922

Which of the following is a potential remedy for multicollinearity?

- A) Take first differences of the dependent variable.
- B) Add dummy variables to the regression.
- C) Omit one or more of the collinear variables.



Explanation

The first differencing is not a remedy for the collinearity, nor is the inclusion of dummy variables. The best potential remedy is to attempt to eliminate highly correlated variables.

(Module 1.3, LOS 1.i)

Peter Pun, an enrolled candidate for the CFA Level II examination, has decided to perform a calendar test to examine whether there is any abnormal return associated with investments

and disinvestments made in blue-chip stocks on particular days of the week. As a proxy for blue-chips, he has decided to use the S&P 500 Index. The analysis will involve the use of dummy variables and is based on the past 780 trading days. Here are selected findings of his study:

RSS 0.0039

SSE 0.9534

SST 0.9573

R-squared 0.004

SEE 0.035

Jessica Jones, CFA, a friend of Peter, overhears that he is interested in regression analysis and warns him that whenever heteroskedasticity is present in multiple regression, it could undermine the regression results. She mentions that one easy way to spot conditional heteroskedasticity is through a scatter plot, but she adds that there is a more formal test. Unfortunately, she can't quite remember its name. Jessica believes that heteroskedasticity can be rectified using White-corrected standard errors. Her son Jonathan who has also taken part in the discussion, hears this comment and argues that White corrections would typically reduce the number of Type I errors in financial data.

Question #139 - 142 of 164

Question ID: 1472057

What is *most likely* represented by the Y intercept of the regression?

- A) The intercept is not a driver of returns, only the independent variables. 
- B) The return on a particular trading day. 
- C) The drift of a random walk. 

Explanation

The omitted variable is represented by the intercept. So, if we have four variables to represent Monday through Thursday, the intercept would represent returns on Friday. Remember when we want to distinguish between "n" classes we always use one less dummy variable the number of classes ($n - 1$).

(Module 1.4, LOS 1.I)

Question #140 - 142 of 164

Question ID: 1479957

What can be said of the overall explanatory power of the model at the 5% significance?

A) There is no value to calendar trading. 

The coefficient of determination for the above regression is significantly higher

B) than the standard error of the estimate, and therefore there is value to calendar trading. 

C) There is value to calendar trading. 

Explanation

This question calls for a computation of the F -stat for all independent variables jointly. $F = (0.0039 / 4) / (0.9534 / (780 - 4 - 1)) = 0.79$. The critical F is somewhere between 2.37 and 2.45 so we fail to reject the null that the coefficient are equal to zero.

(Module 1.2, LOS 1.e)

Question #141 - 142 of 164

Question ID: 1472059

The test mentioned by Jessica is known as the:

A) Durbin-Watson, which is a two-tailed test. 

B) Breusch-Pagan, which is a one-tailed test. 

C) Breusch-Pagan, which is a two-tailed test. 

Explanation

The Breusch-Pagan is used to detect conditional heteroskedasticity and it is a one-tailed test. This is because we are only concerned about large values in the residuals coefficient of determination.

(Module 1.3, LOS 1.h)

Question #142 - 142 of 164

Question ID: 1479958

Are Jessica and her son Jonathan correct in terms of the method used to correct for heteroskedasticity and the likely effects?

A) Neither is correct.



B) Both are correct.



C) One is correct.



Explanation

Jessica is correct. White-corrected standard errors are also known as robust standard errors. Jonathan is correct because for financial data, generally, White-corrected errors are higher than the biased errors leading to lower computed t-statistics and, therefore, less frequent rejection of the null hypothesis (remember incorrectly rejecting a true null is Type I error).

(Module 1.3, LOS 1.h)

Question #143 of 164

Question ID: 1471891

Which of the following statements regarding the results of a regression analysis is *least* accurate? The:

A) slope coefficient in a multiple regression is the value of the dependent variable for a given value of the independent variable.



B) slope coefficient in a multiple regression is the change in the dependent variable for a one-unit change in the independent variable, holding all other variables constant.



C) slope coefficients in the multiple regression are referred to as partial betas.



Explanation

The slope coefficient is the change in the dependent variable for a one-unit change in the independent variable.

(Module 1.1, LOS 1.b)

Question #144 of 164

Question ID: 1479906

Which of the following is *least likely* a method used to detect heteroskedasticity?

A) Scatter plot.



B) Breusch-Godfrey test.



C) Breusch-Pagan test.



Explanation

The Breusch-Godfrey test is used to detect serial correlation. The Breusch-Pagan test is a formal test used to detect heteroskedasticity while a scatter plot can give visual clues about presence of heteroscedasticity.

(Module 1.3, LOS 1.h)

Question #145 of 164

Question ID: 1472009

Which of the following statements regarding heteroskedasticity is *least* accurate?

- A) Conditional heteroskedasticity can be detected using the Breusch-Pagan chi-square statistic. 
- B) When not related to independent variables, heteroskedasticity does not pose any major problems with the regression. 
- C) Heteroskedasticity only occurs in cross-sectional regressions. 

Explanation

If there are shifting regimes in a time-series (e.g., change in regulation, economic environment), it is possible to have heteroskedasticity in a time-series. Unconditional heteroskedasticity occurs when the heteroskedasticity is not related to the level of the independent variables. Unconditional heteroskedasticity causes no major problems with the regression. Breusch-Pagan statistic has a chi-square distribution and can be used to detect conditional heteroskedasticity.

(Module 1.3, LOS 1.h)

Question #146 of 164

Question ID: 1472007

An analyst is trying to estimate the beta for a fund. The analyst estimates a regression equation in which the fund returns are the dependent variable and the Wilshire 5000 is the independent variable, using monthly data over the past five years. The analyst finds that the correlation between the square of the residuals of the regression and the Wilshire 5000 is 0.2. Which of the following is *most* accurate, assuming a 0.05 level of significance? There is:

- A) no evidence that there is conditional heteroskedasticity or serial correlation in the regression equation. 
- B) evidence of conditional heteroskedasticity but not serial correlation in the regression equation. 

- c) evidence of serial correlation but not conditional heteroskedasticity in the regression equation.



Explanation

The test for conditional heteroskedasticity involves regressing the square of the residuals on the independent variables of the regression and creating a test statistic that is $n \times R^2$, where n is the number of observations and R^2 is from the squared-residual regression. The test statistic is distributed with a chi-squared distribution with the number of degrees of freedom equal to the number of independent variables. For a single variable, the R^2 will be equal to the square of the correlation; so in this case, the test statistic is $60 \times 0.2^2 = 2.4$, which is less than the chi-squared value (with one degree of freedom) of 3.84 for a p-value of 0.05. There is no indication about serial correlation.

(Module 1.3, LOS 1.h)

Quin Tan Liu, CFA, is looking at the retail property sector for her manager. She is undertaking a top down review as she feels this is the best way to analyze the industry segment. To predict U.S. property starts (housing), she has used regression analysis.

Liu included the following variables in her analysis:

- Average nominal interest rates during each year (as a decimal)
- Annual GDP per capita in \$'000

Given these variables the following output was generated from 30 years of data:

Exhibit 1 – Results from Regressing Housing Starts (in Millions) on Interest Rates and GDP Per Capita

		Coefficient	Standard Error	T-statistic
Intercept		0.42		3.1
Interest rate		-1.0		-2.0
GDP per capita		0.03		0.7
ANOVA	df	SS	MSS	F
Regression	2	3.896	1.948	21.644
Residual	27	2.431	0.090	
Total	29	6.327		
Observations	30			
Durbin-Watson	1.22			

Exhibit 2 - Critical Values for F-Distribution at 5% Level of Significance

Degrees of Freedom for the Denominator	Degrees of Freedom (df) for the Numerator		
	1	2	3
26	4.23	3.37	2.98
27	4.21	3.35	2.96
28	4.20	3.34	2.95
29	4.18	3.33	2.93
30	4.17	3.32	2.92
31	4.16	3.31	2.91
32	4.15	3.30	2.90

The following variable estimates have been made for 20X7:

GDP per capita = \$46,700

Interest rate = 7%

Question #147 - 149 of 164

Question ID: 1479885

Using the regression model represented in Exhibit 1, what is the predicted number of housing starts for 20X7?

- A) 1,394
- B) 1,394,420
- C) 1,751,000

**Explanation**

Housing starts = $0.42 - (1 \times 0.07) + (0.03 \times 46.7) = 1.751$ million

(Module 1.2, LOS 1.f)

Question #148 - 149 of 164

Question ID: 1479886

Which of the following statements *best* describes the explanatory power of the estimated regression?

- A) The residual standard error of only 0.3 indicates that the regression equation is a good fit for the sample data. 
- B) The independent variables explain 61.58% of the variation in housing starts. 
- C) The large F -statistic indicates that both independent variables help explain changes in housing starts. 

Explanation

The coefficient of determination is the statistic used to identify explanatory power. This can be calculated from the ANOVA table as $3.896/6.327 \times 100 = 61.58\%$.

The residual standard error of 0.3 indicates that the standard deviation of the residuals is 0.3 million housing starts. Without knowledge of the data for the dependent variable it is not possible to assess whether this is a small or a large error.

The F -statistic does not enable us to conclude on both independent variables. It only allows us to reject the hypothesis that all regression coefficients are zero and accept the hypothesis that at least one isn't.

(Module 1.2, LOS 1.d)

(Module 1.2, LOS 1.e)

Question #149 - 149 of 164

Question ID: 1479887

Which of the following is the least appropriate statement in relation to R-square and adjusted R-square:

- A) Adjusted R-square decreases when the added independent variable adds little value to the regression model. 
- B) Adjusted R-square is a value between 0 and 1 and can be interpreted as a percentage. 
- C) R-square typically increases when new independent variables are added to the regression regardless of their explanatory power. 

Explanation

Adjusted R-square can be negative for a large number of independent variables that have no explanatory power. The other two statements are correct.

(Module 1.2, LOS 1.d)

William Brent, CFA, is the chief financial officer for Mega Flowers, one of the largest producers of flowers and bedding plants in the Western United States. Mega Flowers grows

its plants in three large nursery facilities located in California. Its products are sold in its company-owned retail nurseries as well as in large, home and garden "super centers". For its retail stores, Mega Flowers has designed and implemented marketing plans each season that are aimed at its consumers in order to generate additional sales for certain high-margin products. To fully implement the marketing plan, additional contract salespeople are seasonally employed.

For the past several years, these marketing plans seemed to be successful, providing a significant boost in sales to those specific products highlighted by the marketing efforts. However, for the past year, revenues have been flat, even though marketing expenditures increased slightly. Brent is concerned that the expensive seasonal marketing campaigns are simply no longer generating the desired returns, and should either be significantly modified or eliminated altogether. He proposes that the company hire additional, permanent salespeople to focus on selling Mega Flowers' high-margin products all year long. The chief operating officer, David Johnson, disagrees with Brent. He believes that although last year's results were disappointing, the marketing campaign has demonstrated impressive results for the past five years, and should be continued. His belief is that the prior years' performance can be used as a gauge for future results, and that a simple increase in the sales force will not bring about the desired results.

Brent gathers information regarding quarterly sales revenue and marketing expenditures for the past five years. Based upon historical data, Brent derives the following regression equation for Mega Flowers (stated in millions of dollars):

$$\text{Expected Sales} = 12.6 + 1.6 (\text{Marketing Expenditures}) + 1.2 (\# \text{ of Salespeople})$$

Brent shows the equation to Johnson and tells him, "This equation shows that a \$1 million increase in marketing expenditures will increase the independent variable by \$1 .6 million, all other factors being equal." Johnson replies , "It also appears that sales will equal \$12.6 million if all independent variables are equal to zero."

Question #150 - 155 of 164

Question ID: 1471900

In regard to their conversation about the regression equation:

- A) Brent's statement is correct; Johnson's statement is incorrect.
- B) Brent's statement is incorrect; Johnson's statement is correct.
- C) Brent's statement is correct; Johnson's statement is correct.



Explanation

Expected sales is the dependent variable in the equation, while expenditures for marketing and salespeople are the independent variables. Therefore, a \$1 million increase in marketing expenditures will increase the *dependent* variable (expected sales) by \$1.6 million. Brent's statement is incorrect.

Johnson's statement is correct. 12.6 is the intercept in the equation, which means that if all independent variables are equal to zero, expected sales will be \$12.6 million.

(Module 1.1, LOS 1.b)

Question #151 - 155 of 164

Question ID: 1471901

Using data from the past 20 quarters, Brent calculates the t-statistic for marketing expenditures to be 3.68 and the t-statistic for salespeople at 2.19. At a 5% significance level, the two-tailed critical values are $t_c = \pm 2.127$. This *most likely* indicates that:

- A) the *t*-statistic has 18 degrees of freedom. 
- B) the null hypothesis should not be rejected. 
- C) both independent variables are statistically significant. 

Explanation

Using a 5% significance level with degrees of freedom (df) of 17 ($20 - 2 - 1$), both independent variables are significant and contribute to the level of expected sales.

(Module 1.1, LOS 1.b)

Question #152 - 155 of 164

Question ID: 1471902

Brent calculated that the sum of squared errors (SSE) for the variables is 267. The mean squared error (MSE) would be:

- A) 14.831. 
- B) 14.055. 
- C) 15.706. 

Explanation

The MSE is calculated as $SSE / (n - k - 1)$. Recall that there are twenty observations and two independent variables. Therefore, the MSE in this instance $[267 / (20 - 2 - 1)] = 15.706$.

(Module 1.1, LOS 1.b)

Question #153 - 155 of 164

Question ID: 1471903

Brent is trying to explain the concept of the standard error of estimate (SEE) to Johnson. In his explanation, Brent makes three points about the SEE:

- Point 1: The SEE is the standard deviation of the differences between the estimated values for the independent variables and the actual observations for the independent variable.
- Point 2: Any violation of the basic assumptions of a multiple regression model is going to affect the SEE.
- Point 3: If there is a strong relationship between the variables and the SSE is small, the individual estimation errors will also be small.

How many of Brent's points are *most* accurate?

A) All 3 of Brent's points are correct.



B) 1 of Brent's points are correct.



C) 2 of Brent's points are correct.

**Explanation**

The statements that if there is a strong relationship between the variables and the SSE is small, the individual estimation errors will also be small, and also that any violation of the basic assumptions of a multiple regression model is going to affect the SEE are both correct.

The SEE is the standard deviation of the differences between the estimated values for the dependent variables (not independent) and the actual observations for the dependent variable. Brent's Point 1 is incorrect.

Therefore, 2 of Brent's points are correct.

(Module 1.1, LOS 1.b)

Question #154 - 155 of 164

Question ID: 1471904

Assuming that next year's marketing expenditures are \$3,500,000 and there are five salespeople, predicted sales for Mega Flowers should will be:

A) \$24,200,000.



B) \$11,600,000.



C) \$24,000,000.



Explanation

Using the information provided, expected sales equals $12.6 + (1.6 \times 3.5) + (1.2 \times 5) = \24.2 million. Remember to check the details - i.e. this equation is denominated in millions of dollars.

(Module 1.1, LOS 1.b)

Question #155 - 155 of 164

Question ID: 1471905

Brent would like to further investigate whether at least one of the independent variables can explain a significant portion of the variation of the dependent variable. Which of the following methods would be best for Brent to use?

A) An ANOVA table.



B) The multiple coefficient of determination.



C) The *F*-statistic.



Explanation

To determine whether at least one of the coefficients is statistically significant, the calculated *F*-statistic is compared with the critical *F*-value at the appropriate level of significance.

(Module 1.1, LOS 1.b)

Question #156 of 164

Question ID: 1471881

Which of the following statements *most* accurately interprets the following regression results at the given significance level?

Variable	p-value
Intercept	0.0201
X1	0.0284
X2	0.0310
X3	0.0143

A) The variables X1 and X2 are statistically significantly different from zero at the 2% significance level.



B) The variable X3 is statistically significantly different from zero at the 2% significance level.



C) The variable X2 is statistically significantly different from zero at the 3% significance level.



Explanation

The p -value is the smallest level of significance for which the null hypothesis can be rejected. An independent variable is significant if the p -value is less than the stated significance level. In this example, X3 is the variable that has a p -value less than the stated significance level.

(Module 1.1, LOS 1.b)

Lynn Carter, CFA, is an analyst in the research department for Smith Brothers in New York. She follows several industries, as well as the top companies in each industry. She provides research materials for both the equity traders for Smith Brothers as well as their retail customers. She routinely performs regression analysis on those companies that she follows to identify any emerging trends that could affect investment decisions.

Due to recent layoffs at the company, there has been some consolidation in the research department. Two research analysts have been laid off, and their workload will now be distributed among the remaining four analysts. In addition to her current workload, Carter will now be responsible for providing research on the airline industry. Pinnacle Airlines, a leader in the industry, represents a large holding in Smith Brothers' portfolio. Looking back over past research on Pinnacle, Carter recognizes that the company historically has been a strong performer in what is considered to be a very competitive industry. The stock price over the last 52-week period has outperformed that of other industry leaders, although Pinnacle's net income has remained flat. Carter wonders if the stock price of Pinnacle has become overvalued relative to its peer group in the market, and wants to determine if the timing is right for Smith Brothers to decrease its position in Pinnacle.

Carter decides to run a regression analysis, using the monthly returns of Pinnacle stock as the dependent variable and monthly returns of the airlines industry as the independent variable.

Analysis of Variance Table (ANOVA)			
Source	df (Degrees of Freedom)	SS (Sum of Squares)	Mean Square (SS/df)
Regression	1	3,257 (RSS)	3,257 (MSR)

Error	8	298 (SSE)	37.25 (MSE)
Total	9	3,555 (SS Total)	

Question #157 - 160 of 164

Question ID: 1479893

Which of the following is *least likely* to be an assumption regarding linear regression?

- A) The independent variable is correlated with the residuals. 
- B) The variance of the residuals is constant. 
- C) A linear relationship exists between the dependent and independent variables. 

Explanation

Although the linear regression model is fairly insensitive to minor deviations from any of these assumptions, the independent variable is typically uncorrelated with the residuals.

(Module 1.1, LOS 1.c)

Question #158 - 160 of 164

Question ID: 1471957

Based upon the information presented in the ANOVA table, what is the coefficient of determination?

- A) 0.916, indicating that the variability of industry returns explains about 91.6% of the variability of company returns. 
- B) 0.084, indicating that the variability of industry returns explains about 8.4% of the variability of company returns. 
- C) 0.839, indicating that company returns explain about 83.9% of the variability of industry returns. 

Explanation

The coefficient of determination (R^2) is the percentage of the total variation in the dependent variable explained by the independent variable.

The $R^2 = (RSS / SS) \text{ Total} = (3,257 / 3,555) = 0.916$. This means that the variation of independent variable (the airline industry) explains 91.6% of the variations in the dependent variable (Pinnacle stock).

(Module 1.2, LOS 1.d)

Question #159 - 160 of 164

Question ID: 1479894

Based upon her analysis, Carter has derived the following regression equation: $\hat{Y} = 1.75 + 3.25X_1$. The predicted value of the Y variable equals 50.50, if the:

- A) predicted value of the independent variable equals 15.
- B) predicted value of the dependent variable equals 15.
- C) coefficient of the determination equals 15.



Explanation

Note that the easiest way to answer this question is to plug numbers into the equation.

The predicted value for $Y = 1.75 + 3.25(15) = 50.50$.

The variable X_1 represents the independent variable.

(Module 1.2, LOS 1.f)

Question #160 - 160 of 164

Question ID: 1479895

Carter realizes that although regression analysis is a useful tool when analyzing investments, there are certain limitations. Carter made a list of points describing limitations that Smith Brothers equity traders should be aware of when applying her research to their investment decisions.

- Point 1: Regression residuals may be homoskedastic.
- Point 2: Data from regression relationships tends to exhibit parameter instability.
- Point 3: Regression residuals may exhibit autocorrelation.
- Point 4: The variance of the error term may change with one or more independent variables.

When reviewing Carter's list, one of the Smith Brothers' equity traders points out that not all of the points describe regression analysis limitations. Which of Carter's points most accurately describes the limitations to regression analysis?

- A) Points 1, 3, and 4.
- B) Points 2, 3, and 4.
- C) Points 1, 2, and 3.



Explanation

One of the basic assumptions of regression analysis is that the variance of the error terms is constant, or homoskedastic. Any violation of this assumption is called heteroskedasticity. Therefore, Point 1 is incorrect, but Point 4 is correct because it describes conditional heteroskedasticity, which results in unreliable estimates of standard errors. Points 2 and 3 also describe limitations of regression analysis.

(Module 1.1, LOS 1.c)

Using a recent analysis of salaries (in \$1,000) of financial analysts, Timbadia runs a regression of salaries on education, experience, and gender. (Gender equals one for men and zero for women.) The regression results from a sample of 230 financial analysts are presented below, with *t*-statistics in parenthesis.

$$\text{Salary} = 34.98 + 1.2 \text{ Education} + 0.5 \text{ Experience} + 6.3 \text{ Gender}$$

(29.11) (8.93) (2.98) (1.58)

Timbadia also runs a multiple regression to gain a better understanding of the relationship between lumber sales, housing starts, and commercial construction. The regression uses a large data set of lumber sales as the dependent variable with housing starts and commercial construction as the independent variables. The results of the regression are:

	Coefficient	Standard Error	t-statistics
Intercept	5.337	1.71	3.14
Housing starts	0.76	0.09	8.44
Commercial construction	1.25	0.33	3.78

Finally, Timbadia runs a regression between the returns on a stock and its industry index with the following results:

	Coefficient	Standard Error
Intercept	2.1	2.01
Industry index	1.9	0.31

- Standard error of estimate = 15.1
- Correlation coefficient = 0.849

What is the expected salary (in \$1,000) of a woman with 16 years of education and 10 years of experience?

A) 65.48.



B) 54.98.



C) 59.18.



Explanation

$$34.98 + 1.2(16) + 0.5(10) = 59.18$$

(Module 1.2, LOS 1.f)

Question #162 - 163 of 164

Question ID: 1479890

If the return on the industry index is 4%, the stock's expected return would be:

A) 7.6%.



B) 11.2%.



C) 9.7%.



Explanation

$$Y = b_0 + bX_1$$

$$Y = 2.1 + 1.9(4) = 9.7\%$$

(Module 1.2, LOS 1.f)

Question #163 - 163 of 164

Question ID: 1479891

The percentage of the variation in the stock return explained by the variation in the industry index return is *closest* to:

A) 84.9%.



B) 72.1%.



C) 63.2%.



Explanation

The coefficient of determination, R^2 , is the square the correlation coefficient. $0.849^2 = 0.721$.

(Module 1.2, LOS 1.d)

Question #164 of 164

Question ID: 1479949

Suppose the analyst wants to add a dummy variable for whether a person has a business college degree and an engineering degree. What is the *CORRECT* representation if a person has both degrees?

	<u>Business</u> <u>Degree Dummy</u> <u>Variable</u>	<u>Engineering</u> <u>Degree Dummy</u> <u>Variable</u>	
A)	1	1	
B)	0	1	
C)	0	0	

Explanation

Assigning a zero to both categories is appropriate for someone with neither degree. Assigning one to the business category and zero to the engineering category is appropriate for someone with only a business degree. Assigning zero to the business category and one to the engineering category is appropriate for someone with only an engineering degree. Assigning a one to both categories is correct because it reflects the possession of both degrees.

(Module 1.4, LOS 1.I)